MILESTONE 1 · ANALYTICAL REPORT

Formula 1 Podium Prediction Companion

Predicting, for every driver in a Grand Prix, whether they finish on the podium, using only what is known after qualifying and before the race starts.

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Github Link: https://github.com/George140105/ACL2_Milestone1_Team29

Kaggle Notebook: https://www.kaggle.com/code/georgeamgad11/acl-milestone-1-team29

1 Summary and the big picture

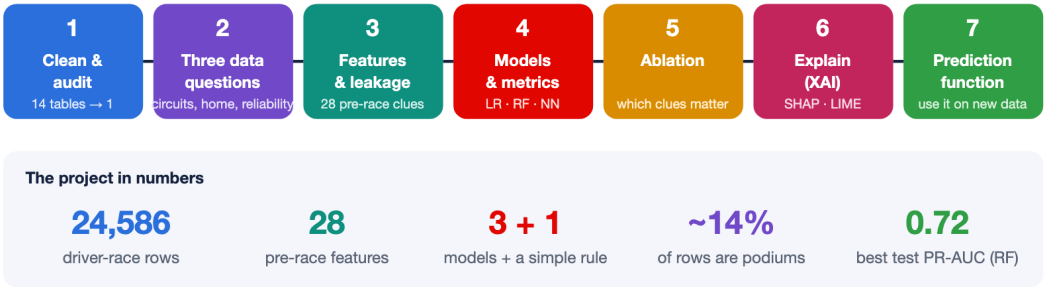


Diagram 1. *The seven-step pipeline, one step per notebook section, and the project in numbers.*

The prediction cutoff

Every design decision follows from the moment of prediction: qualifying has finished and the grid is fixed, but the race has not started. Inputs must exist at that moment; anything produced by the race itself is a future value and would be data leakage.

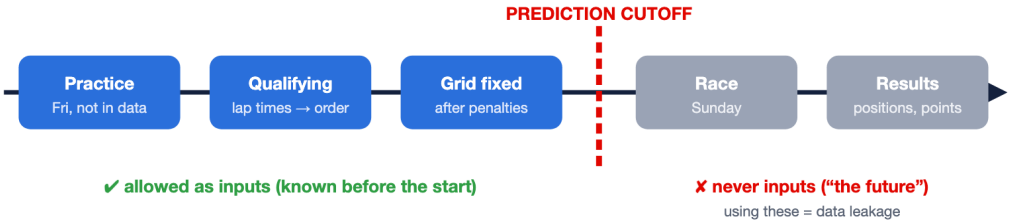


Diagram 2. *The race weekend and the prediction cutoff. Information to the left of the red line may be used as model input; information to the right never is.*

Overview of the results

The task is to predict, for one driver in one Grand Prix, whether the driver finishes on the podium (positionOrder 3 or better). The prediction is made after qualifying and before the race, so every input has to be known by then. We use the Ergast data (1950 to 2024, 14 tables), clean it into one row per driver and race (24,586 rows), build 28 pre-race features in seven groups, and compare logistic regression, a random forest and a shallow neural network on a chronological split: training 2003 to 2019, validation 2020 and 2021, test 2022 onward.

The main results are these. A simple rule (podium if the driver starts in the top three) already reaches a test PR-AUC of 0.647. The three models reach 0.698 to 0.720 and a test ROC-AUC of 0.926 to 0.938, and they are statistically tied with each other. Only the random forest's ROC-AUC advantage over the logistic baseline is reliable. Starting position and qualifying dominate every explanation, followed by recent driver form, and the features pass a future-invariance leakage test. The three data-engineering questions give a mixed picture: circuits cannot be separated on front-row conversion, there is no detectable home-race effect once the grid slot is controlled, and mechanical retirements fell from 8.9% to 5.4% of starts between 2014 to 2021 and 2022 to 2024.

Section 9 lists the limits that matter most: a small validation set (39 races), qualifying data that only exist in full from 2003, results that are tied across models, and explanations that describe what the models use and not what causes podiums.

2 Data audit, cleaning and splits

WHAT	HOW	WHY
Find and fix placeholder missing values, wrong data types, duplicate driver-race rows and broken links between tables.	Automated checks and plots before and after cleaning, and five logged rules (R1 to R5), each with the number of rows it touches.	Each row is one prediction. A duplicate would count a driver twice; an untyped value cannot be used; an orphan key breaks the joins.

2.1 Audit

All 14 CSV files were loaded unchanged and every cleaning step was applied to copies. The files use the literal string \N as a missing-value sentinel, which we converted to proper missing values before any typing. The largest sentinel shares are results.position (40.9%, retired drivers), qualifying q3 (65.0%) and q2 (44.1%), while q1 is missing in only 1.5% of rows. Numeric columns that arrived as text because of the sentinel became numbers, qualifying times were parsed from m:ss.mmm text to seconds, dates (race dates, birth dates) became datetimes, and durations were typed as well: pit-stop durations (text such as 22.552 or 1:01.235), lap times and race times became seconds and race start times became time values. The durations are post-race fields used only for exploration, never as features.

The raw grid column holds 0 in 1,638 of 26,759 rows (6.1%). Almost all of these are drivers who never started (did not qualify 1,013, did not pre-qualify 331, withdrew 164); only a small number are genuine pit-lane starts (87 after cleaning; none of the 73 since 2003 reached the podium), which is why rules R2 and R4 treat the two cases differently.

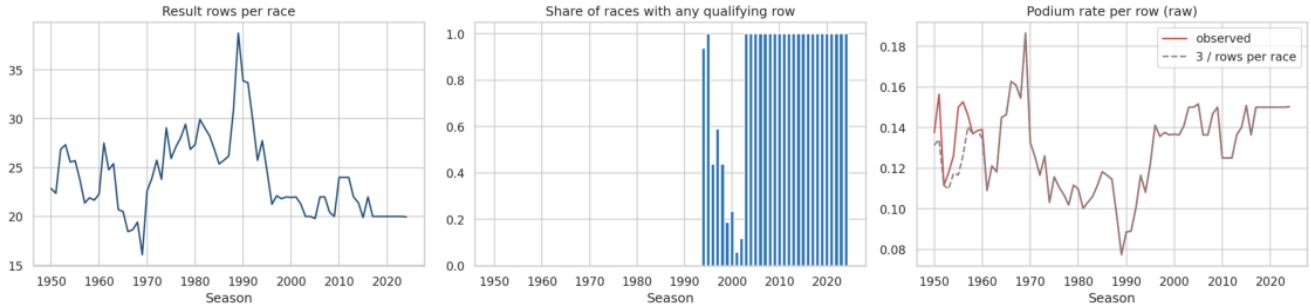


Figure 1. *Raw exploratory view by season: entries per race (left), share of races with any qualifying row (middle) and raw podium rate per row against the 3-over-field-size expectation (right).*

WHAT THE CHART SHOWS

The left panel shows the number of result rows per race. Fields of 25 to 39 entries between the 1950s and the 1990s include many cars that never started, while modern races have about 20. The middle panel shows that no race has a qualifying record before 1994, so qualifying features cannot exist for most of the history. The right panel shows that the raw podium rate closely follows 3 divided by the field size, so the base rate depends on how many cars enter. This is why non-starters are removed (rule R2) and the podium rate is reported per split.

Every table has a unique, non-null primary key. The natural key (raceId, driverId) is not unique in results: 91 rows are duplicates, which is the grain problem we fix below. All 23 foreign-key relations reconcile, with zero orphan references. The target uses positionOrder because it is the only position column filled for every row and it already reflects the final classification, including post-race disqualifications. The podium rate per row moves with field size (3 podium places divided by about 20 to 28 entries), so we later report it by split rather than as one number.

2.2 Cleaning rules and grain

The target grain is one row per (raceId, driverId). Five rules produce it, each with the rows it affects.

Table 1. *Cleaning rules (26,759 raw result rows become 24,586).*

Rule	What it does	Rows	Why
R1	Drop the Indianapolis 500 races of 1950 to 1960	405 (11 races)	The race counted for the championship but almost no regular F1 drivers entered it. It would inflate race counts for American drivers (distorting the home-race question) and has a different grid structure
R2	Drop non-starters (positionText F or W, or status DNQ and DNPQ)	1,704	They never raced, so no podium prediction applies, and removing them also stops the oversized 1980s and 1990s fields (many cars that never started) from lowering the base rate. 153 of them had a grid slot (e.g. withdrawals), 47 of those from 2003 on. Whether a qualified driver will start is not known at the prediction cutoff, so for these 47 rows the rule uses information from just after the cutoff (for example Vettel, Bahrain 2016, who failed before the start from P3). We accept this because the target is undefined for a car that never races and all 47 rows are non-podiums
R3	Resolve shared drives: 61 pairs, 125 rows. Keep the row with a non-zero grid, then the lowest grid, then the lowest resultId	64 dropped	One driver, one race, one row. In 17 pairs the dropped row was a podium and the kept row was not, so the rule is documented and deterministic
R4	Raw grid 0 means pit-lane start. Set grid_clean to the last slot plus one and add pit_lane_start	0 dropped, 87 flagged	Keeps grid monotone (a raw 0 would look like pole position)
R5	Qualifying laps above 1.5 times the race median become missing	2 laps	One Q1 lap of 16:42.640 (1995 Japanese GP) is a data error

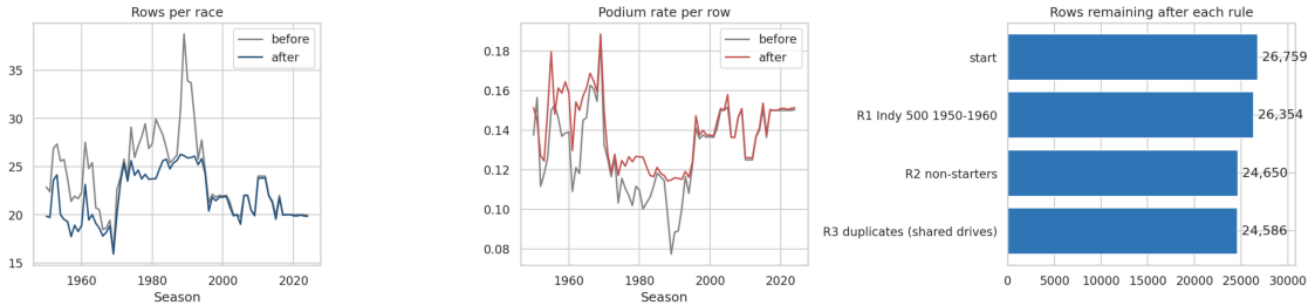


Figure 2. *Effect of cleaning: entries per race (left) and podium rate per row (middle) before and after cleaning, and rows remaining after each rule (right).*

WHAT THE CHART SHOWS

Removing drivers who never started brings the field size down to the actual starters, so the 1980s and 1990s peaks in the left panel disappear. As a result, the podium rate per row rises in the eras that had oversized fields (middle panel). The right panel traces the row count through each rule, from 26,759 raw rows to 24,586.

After cleaning, (raceId, driverId) is unique and the table (24,586 rows, 40 columns once features are added) is saved as the modeling table. Every race from 2003 onward has exactly three podium rows; the 5 races with a different count are all from the 1950s.

2.3 Qualifying coverage and the split

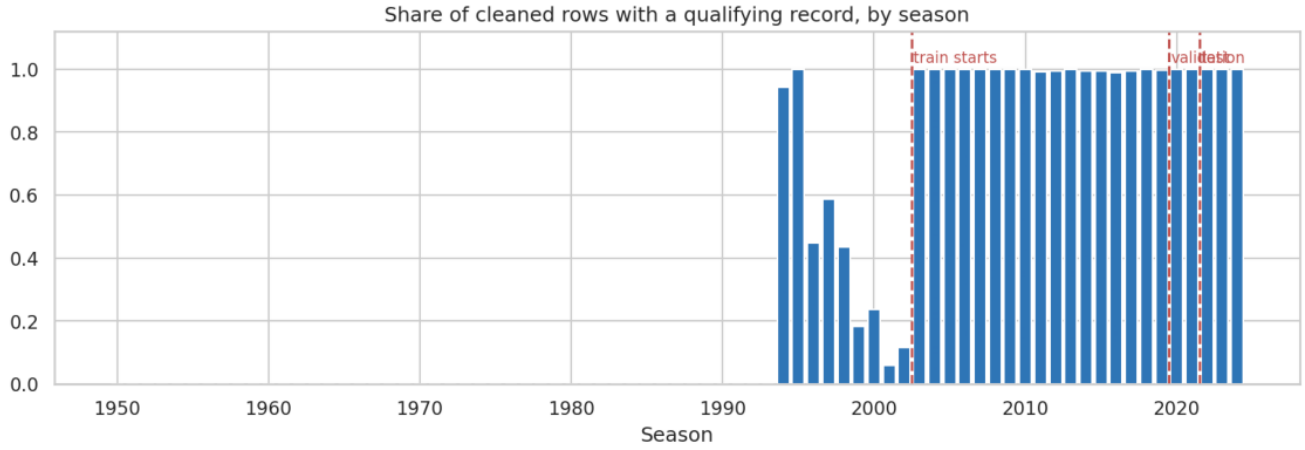


Figure 3. *Share of cleaned rows with a qualifying record, by season. Coverage is 0% before 1994, 46% in 1994 to 2002 and 99.8% from 2003. Dashed lines mark the start of the training, validation and test periods.*

WHAT THE CHART SHOWS

Each bar gives the share of cleaned rows in one season that have a qualifying record. Coverage is zero before 1994, partial until 2002 and 99.8% from 2003 onward, which is the qualifying-coverage drift described in the brief. The dashed lines mark the start of the training, validation and test periods; training begins in the first season with complete coverage.

The drift follows the qualifying formats. Qualifying was not recorded before 1994 and only partly until 2002, the single-lap formats of 2003 to 2005 mostly record only q1, and the three-part knockout (q1, q2, q3) exists from 2006, with drivers knocked out early having no later times. Three options were open for the qualifying features: impute them, fall back to the grid, or restrict the training window. We restrict the window and keep the remaining gaps as missing values with indicator flags. Falling back to the grid would duplicate the grid feature, and imputing 50 years of absent sessions would invent data.

This coverage drift decides where training starts. A model trained on seasons without qualifying data would learn from rows where the most useful input is empty, so training starts in 2003 and the 15,676 earlier rows are kept only as history for the rolling form features. The split is by season rather than random for two reasons. A random split would let the model learn from later seasons and then predict earlier ones, which cannot happen in practice, and it would place drivers from the same race on both sides of the split. Splitting by season keeps each race whole and reproduces real use: learn from past seasons, tune on the next ones and report once on the newest. Validation covers two seasons (39 races, about 115 podiums), the smallest period that still supports threshold tuning. The test seasons (2022 to 2024) follow the ground-effect regulations that no training season saw, which makes the test a deliberately demanding check.

Table 2. *Chronological split.*

Split	Seasons	Rows	Races	Podium rate
History (features only)	1950 to 2002	15,676	686	13.1%
Train	2003 to 2019	6,783	321	14.2%
Validation	2020 to 2021	775	39	15.1%
Test	2022 onward	1,352	68	15.1%

Experiment C checks the choice of 2003: training from 1996, 1980 or 1950 (without qualifying features, which do not exist earlier) changes validation ROC-AUC by at most +0.0013 against 0.9262 for 2003. More old data does not help, and 2003 is the first season of full coverage.

3 Data-engineering questions

Each answer combines several tables. All three use race starts as the grain (one driver in one race) and exclude non-starters through rule R2.

STATISTICS USED IN THIS SECTION

95% interval: the range in which the true value plausibly lies; long intervals mean few observations.

p-value: how surprising the observed result would be if there were no real effect; below 0.05 is read as evidence of an effect.

Chi-square test: do several groups (here, circuits) differ at all? **Fisher exact test:** did a rate change between two groups (here, two eras)?

Odds ratio: how much a factor multiplies the odds of the outcome; 1 means no effect.

3.1 Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Tables joined	Method	Answer
results, races, circuits, and qualifying on (raceId, driverId).	Front row defined twice: by qualifying position and by grid after penalties. Conversion = front-row starters on the podium ÷ front-row starters, 2003 to 2024.	COTA and Monaco (83%) lead and the Nürburgring (56%) trails, but circuits are not statistically distinguishable. The reliable finding is an era-wide rise from 68.3% to 76.2%.

Tables joined: results with races and circuits, and qualifying on (raceId, driverId). Qualifying position and grid position are kept separate: the front row is defined once by qualifying position and once by clean grid, which differs through penalties. The analysis covers 2003 to 2024 because coverage is complete from 2003 (21 of 8,910 rows lack a qualifying position and are dropped) and uses the 24 circuits with at least 8 races. Both definitions are known before the start, so neither leaks.

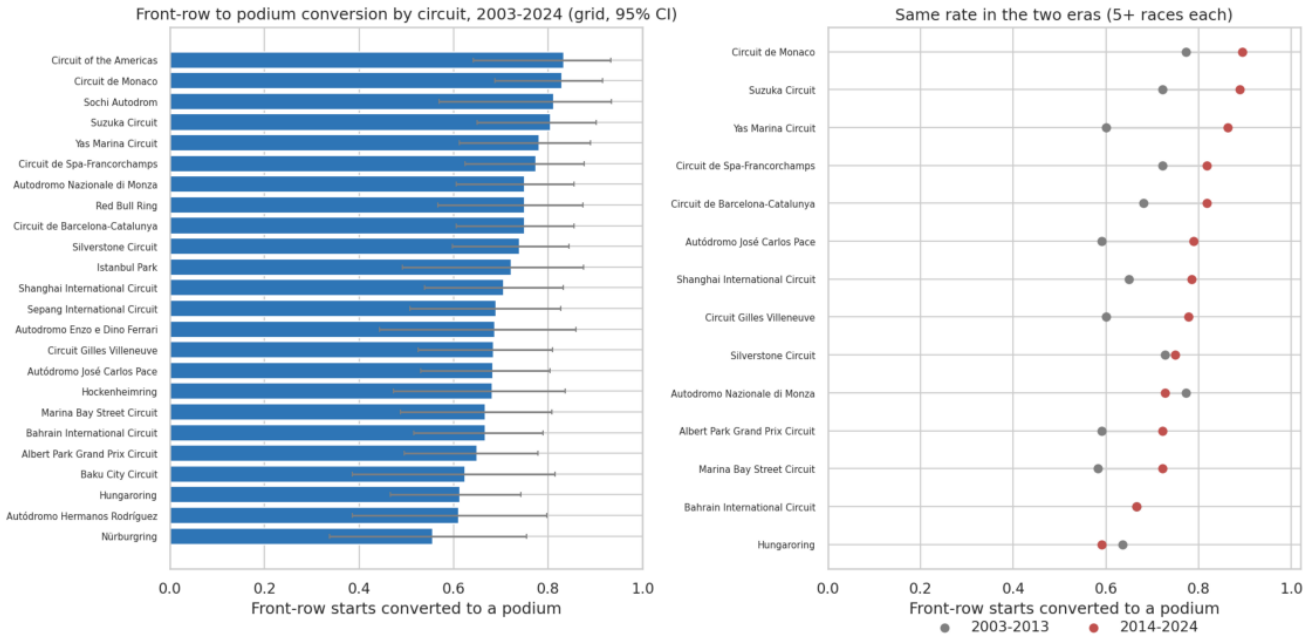


Figure 4. *Left: front-row podium conversion by circuit, 2003 to 2024, grid definition, with 95% intervals. Right: the same rate before and after 2014 for circuits with at least 5 races in each era.*

WHAT THE CHART SHOWS

The left panel ranks circuits by the share of front-row starters (by grid) who finished on the podium, with 95% intervals. The intervals overlap for almost every circuit, which is the visual counterpart of the chi-square result that circuits cannot be separated. The right panel compares the same rate before 2014 (grey) and from 2014 (red) for each circuit. The red point lies further right at 11 of 14 circuits, which points to an era-wide rise rather than a change at particular tracks.

Overall, 72.5% of front-row starters by grid and 71.0% by qualifying position finish on the podium. The two definitions disagree about the front row for only 73 of 8,889 rows (0.8%). By grid, the highest conversion is at the Circuit of the Americas and Monaco (83% each) and at Sochi (81%); the lowest is at the Hungaroring and Mexico City (61% each) and the Nürburgring (56%).

The ranking does change across eras: the rank correlation between the pre-2014 and post-2014 circuit rankings is 0.36 (p = 0.21) by grid and 0.54 (p = 0.047) by qualifying position. Conversion rose at most circuits after 2014, most at Yas Marina (+26 points), Interlagos (+20) and Circuit Gilles Villeneuve (+18), and fell slightly at Monza and the Hungaroring (both -5). However, circuit differences are not statistically distinguishable from sampling noise within either era (chi-square p = 0.93 for 2003 to 2013 and 0.66 for 2014 to 2024 by grid; 0.94 and 0.70 by qualifying), because each circuit contributes only 16 to 46 front-row starts. The reliable finding is the era-wide rise in front-row conversion, not a stable circuit ranking. By grid, conversion rose from 68.3% (272 of 398 front-row starts, 2003 to 2013) to 76.2% (345 of 453, 2014 to 2024), Fisher exact p = 0.011; by qualifying position from 67.6% to 74.0% (p = 0.041). Of the 14 circuits with at least 5 races in both eras, 11 rose and 2 fell (mean change +10.7 points, sign test p = 0.022). The post-2014 period includes the 2022 to 2024 ground-effect seasons, following the brief's wording of before and after the hybrid era.

3.2 Do drivers podium more often when racing at home, after controlling for starting position?

Tables joined	Method	Answer
results, drivers (nationality), races and circuits (country), through a reviewed nationality-to-country map.	Compare home and away drivers who started from the same grid bin; logistic regression with grid bin and decade, standard errors clustered by driver; a within-driver check.	No detectable home advantage: grid-adjusted odds ratio 0.92 (95% CI 0.75 to 1.13). The home flag was therefore not used as a feature.

Tables joined: results with drivers (nationality), races and circuits (country). A start is a home start when the driver's nationality maps to the circuit's country. The mapping was built by hand and reviewed for every nationality in the table; 1,927 of 24,586 rows (7.8%, 499 drivers, 67 circuits) are home starts. Nationalities with no host country in the data (15 nationalities, for example Finnish with 1,155 rows and New Zealander with 386) can never be home. Every other nationality is mapped, and every circuit country has a matching nationality. Control is by grid bin (P1, P2, P3, P4 to 6, P7 to 10, P11 to 15, P16+), which is known before the race; the regression also controls for decade.

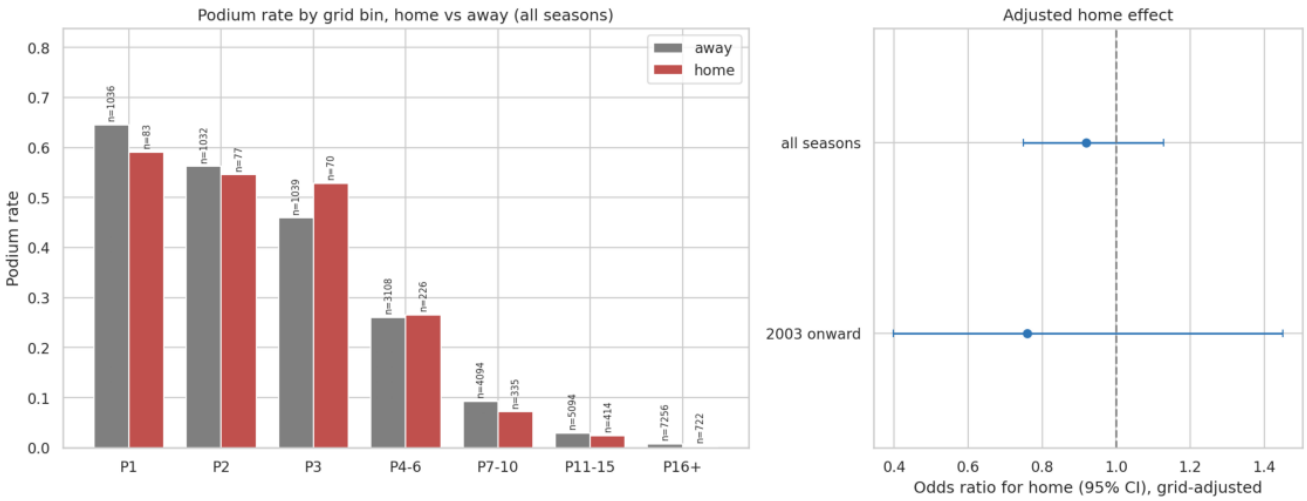


Figure 5. *Left: podium rate by grid bin, home against away starters (counts above the bars). Right: grid-adjusted odds ratio for home racing with 95% intervals; an odds ratio of 1 means no effect.*

WHAT THE CHART SHOWS

Within each grid bin, home starters (red) and away starters (grey) reach almost the same podium rate, so the starting slot, not racing at home, explains the differences. The right panel shows the grid-adjusted odds ratio for racing at home. Both intervals cross the dashed line at 1, so no home effect can be detected in either period.

The raw podium rate is lower at home (11.6%) than away (13.8%), which already argues against a simple home advantage. Comparing drivers from the same grid bins, the adjusted odds ratio is 0.92 (95% CI 0.75 to 1.13, p = 0.42) over all seasons and 0.76 (0.40 to 1.45, p = 0.41) from 2003, with standard errors clustered by driver. A within-driver check on 191 drivers with at least 3 home and 3 away starts gives -0.3 percentage points (bootstrap 95% CI -1.9 to +1.3). The answer is no: we find no detectable home-race advantage after controlling for the grid, and the intervals are wide enough (especially from 2003, with 364 home starts) that a small effect in either direction cannot be excluded. For this reason the home flag was not used as a model feature. Grain and cutoff: one row per start, and the grid bin is fixed before the race, so controlling for it does not use future information. Missingness: drivers from countries that never hosted a race cannot be home by definition; no nationality in the table is left unmapped.

3.3 Which constructors have the highest mechanical-retirement rate in 2014 to 2021 and in 2022 to 2024?

Tables joined	Method	Answer
results, status, races and constructors; renamed teams merged into one lineage.	Every status text grouped as mechanical core, mechanical ambiguous, accident, other or finished. Rate = mechanical retirements per race start.	2014 to 2021: the RB lineage (14.4%), Alpine (13.2%) and McLaren (13.0%) highest. 2022 to 2024: Alpine (10.4%) and Sauber (8.8%). Pooled: 8.9% → 5.4%.

Tables joined: results with status, races and constructors. Every status text was assigned to a group: mechanical core (engine, gearbox, hydraulics, power unit and similar, 361 starts in 2014 to 2024), mechanical ambiguous (tyre, puncture, wing, wheel and similar, 40 starts), accident, other and finished. The headline uses the core definition and the broad definition (core plus ambiguous) is a sensitivity check; their constructor rankings correlate at 0.97 and 0.95. Renamed teams are merged into one lineage (for example Toro Rosso, AlphaTauri and RB), and the rate is mechanical retirements per race start. Starts that ended before the race (rule R2) are excluded, which removes 14 mechanical failures in 2014 to 2024.

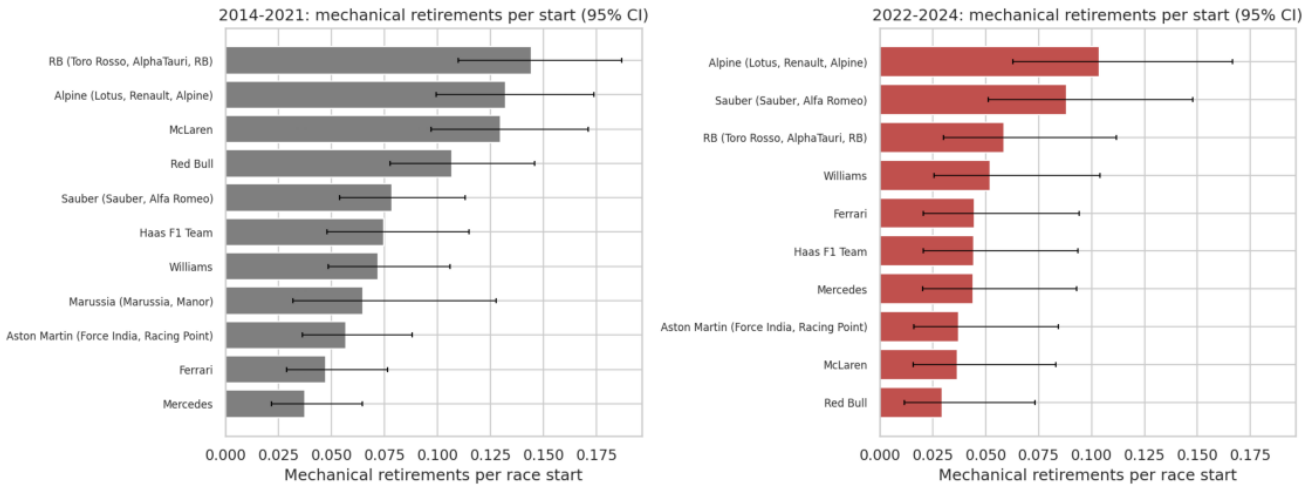


Figure 6. *Mechanical retirements per race start by constructor lineage with 95% intervals, 2014 to 2021 (left) and 2022 to 2024 (right).*

WHAT THE CHART SHOWS

Each bar shows the mechanical retirements per race start for one constructor lineage, with a 95% interval. In 2014 to 2021 the RB, Alpine and McLaren lineages have the highest rates. In 2022 to 2024 each team has only about 135 starts, so the intervals are 6 to 10 percentage points wide and only the top and bottom of the ranking can be told apart.

In 2014 to 2021 the highest rates are Caterham (21.2%, but only 33 starts), the RB lineage (14.4%), the Alpine lineage (13.2%) and McLaren (13.0%); the lowest are Mercedes (3.8%), Ferrari (4.7%) and the Aston Martin lineage (5.7%). In 2022 to 2024 the highest are the Alpine lineage (10.4%), the Sauber lineage (8.8%) and the RB lineage (5.9%); the lowest are Red Bull (2.9%), McLaren and the Aston Martin lineage (3.7% each). Pooled over all constructors the rate fell from 8.9% (288 of 3,246 starts) to 5.4% (73 of 1,352).

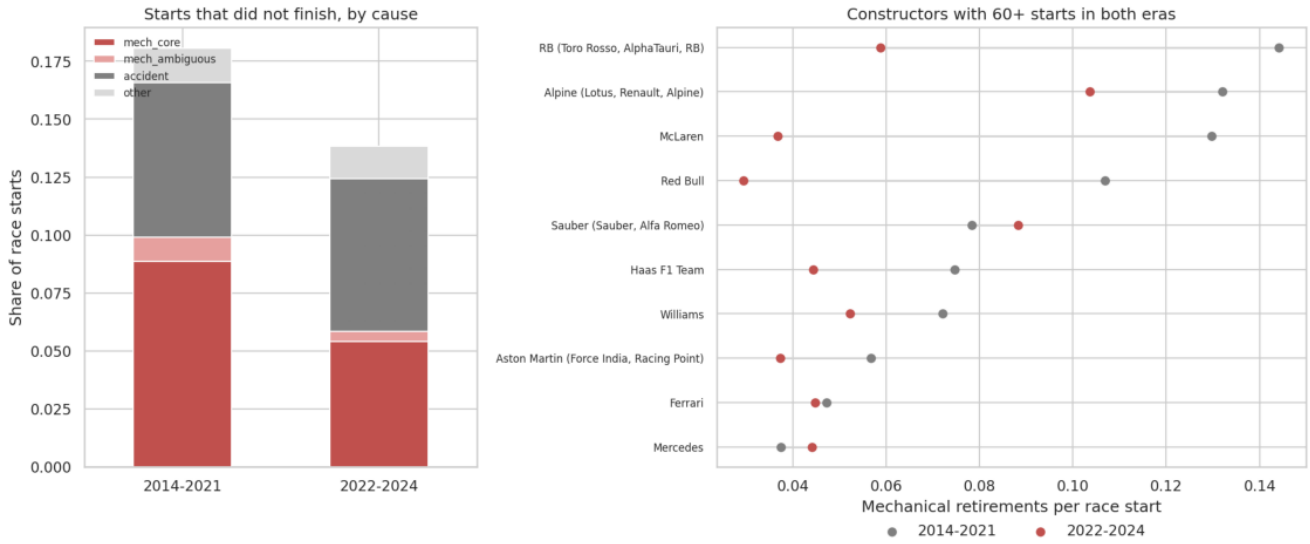


Figure 7. *Left: share of race starts ending in each failure group, by era. Right: core mechanical rate per constructor lineage in both eras (lineages with at least 60 starts in each).*

WHAT THE CHART SHOWS

The left panel splits race starts by how they ended in each era; the mechanical share is clearly smaller in 2022 to 2024. The right panel connects each lineage's mechanical rate in 2014 to 2021 (grey) with its rate in 2022 to 2024 (red). Most points move to the left, meaning better reliability, with the largest improvements at McLaren, the RB lineage and Red Bull.

The biggest improvements are McLaren (-9.3 points), the RB lineage (-8.5) and Red Bull (-7.8); only the Sauber lineage (+1.0) and Mercedes (+0.7) got slightly worse. With about 135 starts per constructor in 2022 to 2024, the intervals are 6 to 10 points wide, so the middle of the 2022 to 2024 ranking is not reliable, while the fall in the pooled rate and the extremes are. Grain, missingness and cutoff: the grain is one start, non-starters are excluded (rule R2), and statuses that fit no group sit in other (49 starts are plain Retired). This is a retrospective analysis, so it may use the status of the race itself; the models only receive the past rate through con_mech_rate_20, computed from earlier races.

Two status values were checked separately: the generic Retired (37 starts in 2014 to 2021, 12 in 2022 to 2024), which does not say why the car stopped, and Fuel leak (1 start), a component failure missing from the core list. Counting both as mechanical raises the pooled rates to 10.0% and 6.4% but leaves the top three of each era unchanged; the rank correlation with the core definition is 0.99 for 2014 to 2021 and 0.94 for 2022 to 2024. The core definition is kept for the headline because it is also the definition behind the con_mech_rate_20 feature.

4 Selected features and why each is valid at the cutoff

The cutoff is the end of qualifying and the announcement of the final grid. A feature is valid if it can be computed from information that exists at that moment. We select 28 features in seven groups; the table gives the definition, the cutoff argument and the strongest single-feature evidence on the training rows (ROC-AUC, direction-adjusted so that 0.5 means no separation).

GRID · 2
Starting slot after penalties; pit-lane flag. The strongest single predictor.

QUALIFYING · 8
Position, percent gap to the session best in Q1, Q2 and Q3, reached Q2/Q3, grid minus qualifying (a penalty signal), missing flag.

DRIVER FORM · 4
Podium rate over the previous 5 and 20 races, mean finish, experience.

TEAM FORM · 4
The same statistics on the team lineage, plus the past mechanical-retirement rate.

STANDINGS · 6
Championship position, wins and points as a share of the leader after the previous round.

CIRCUIT · 2
The driver's own podium rate and number of starts at this circuit before this race.

ERA · 2
Flags for 2003 to 2008 and 2009 to 2013, with 2014 onward as the baseline.

REJECTED
Home flag, raw year, raw points, identity columns, sprint results and every post-race field (Table 4).

Table 3. *The 28 selected features.*

Group	Features	Definition	Valid at the cutoff because	Evidence (AUC)
Grid (2)	grid_clean, pit_lane_start	Starting slot after penalties; pit-lane starters get last slot plus one and a flag	Grid is final after qualifying and penalties, before the start	0.909
Qualifying (8)	qual_pos, q1/q2/ q3_gap_pct, reached_q2, reached_q3, grid_minus_qual, qual_missing	Position, lap-time gap to the session best in percent, whether the driver reached Q2 and Q3, grid minus qualifying position (a penalty signal)	The session ends before the race. Gaps are relative so tracks compare. A missing session is NaN plus a flag, never 0	qual_pos 0.910, q3_gap 0.829, reached_q3 0.792
Driver form (4)	drv_podium_rate_5, drv_podium_rate_20, drv_avg_finish_5, drv_starts_prior	Rolling podium rate and mean finish over the driver's previous 5 and 20 races, and experience	Computed with shift(1) over earlier races only, across the full 1950 onward history	0.878 (rate_20)
Constructor form (4)	con_podium_rate_5, con_podium_rate_20, con_avg_finish_5, con_mech_rate_20	Same rolling statistics on the team lineage (eight renamed-team lineages merging 30 constructor names from 1999, so 166 constructor ids become 154 team keys), plus the past mechanical-retirement rate	Earlier races only. Lineage keys from 1999 merge renamed teams so form carries over	0.882 (rate_20)
Standings (6)	drv_points_share_before, drv_standing_pos_before, drv_wins_before, con_points_share_before, con_standing_pos_before, season_opener	Championship state after the previous round, points as a share of the leader; season_opener flags round 1, when no earlier standing exists	Joined with an as-of merge to an earlier round of the same season; an exact match is disallowed and asserted	0.893 (position)
Circuit (2)	drv_circuit_podium_rate_prior, drv_circuit_starts_prior	The driver's own record at this circuit before this race	Expanding mean over earlier races at the circuit, shifted by one	0.704
Era (2)	era_2003_08, era_2009_13	Regulation-era flags (baseline 2014 onward)	A calendar fact. Replaces the raw year	about 0.51

Window sizes of 5 and 20 races were chosen as a short window that tracks current form and a long one that is stable; they correlate at 0.87 and both are kept because the models are regularised. Experiment F (Section 4.3) rebuilt the form features with windows of 3 and 10 and of 10 and 40 races: validation ROC-AUC changed by -0.0007 and -0.0003, both within noise, so the window choice does not drive the results. Era flags follow the main rule changes in the brief (2009 aerodynamic reset, 2014 hybrid era) with 2014 onward as the baseline, and there is no pre-2003 flag because training starts in 2003. Points are used as a share of the leader because the points scale changed in 2010 (raw points score the same on validation, +0.0007, within noise, so the share is kept for comparability across eras), and eras enter as flags because a raw year would let a model extrapolate a time trend into test seasons it never saw. The era flags are justified by the next figure: the value of a front-row slot is not constant over time.

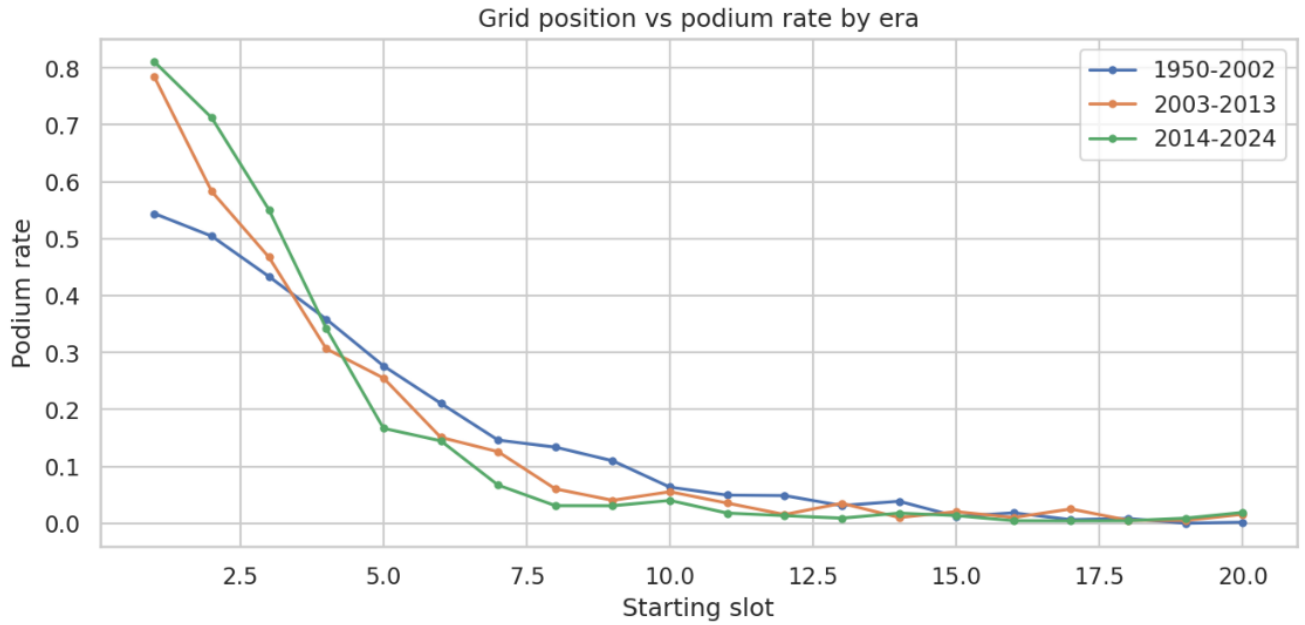


Figure 8. *Podium rate by starting slot in three eras. The pole-position podium rate rises from about 54% (1950 to 2002) to about 78% (2003 to 2013) and 81% (2014 to 2024), and the curve is steeper in the modern eras.*

WHAT THE CHART SHOWS

The chart plots the podium rate for each starting slot in three eras. The podium rate from pole rose from about 54% before 2003 to about 80% in the modern eras, and in every era the rate falls close to zero beyond about fifth place. Because the value of a grid slot depends on the era, the model needs both the grid features and the era flags.

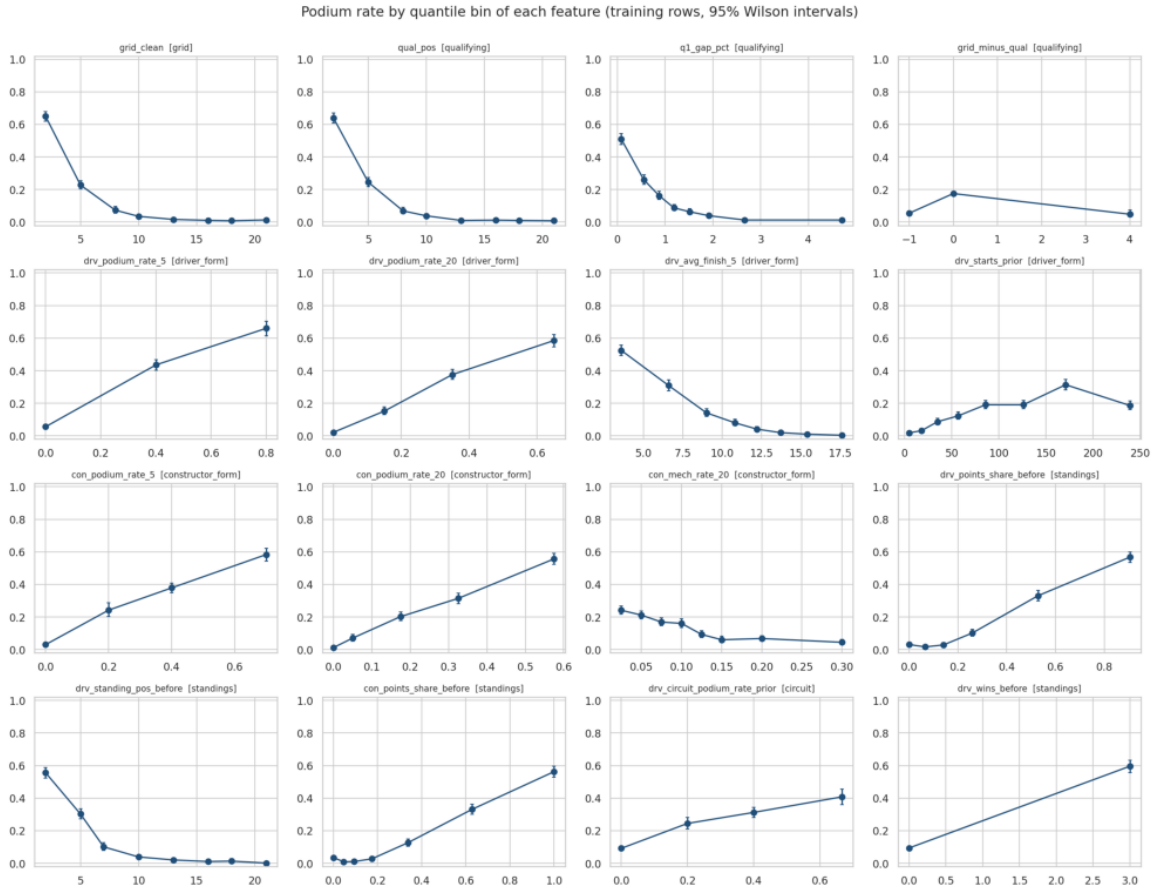


Figure 9. *Podium rate by quantile bin of the main features on the training rows (95% Wilson intervals). Qualifying, grid and form features show clear monotone effects.*

WHAT THE CHART SHOWS

Each panel divides one feature into quantile bins on the training rows and shows the podium rate in each bin with a 95% interval. Grid, qualifying position and the qualifying gaps fall steeply and steadily, while form, points share and wins rise steadily; these are the strongest single predictors. Flat or irregular curves, such as grid minus qualifying and starts at the circuit, mark features with little signal on their own (Table 3a explains why some of them are still kept).

The single-feature screen explains the choice of groups: grid, qualifying, standings and form each separate podium finishers well (AUC 0.88 to 0.91 for the best member), the circuit history is weaker (0.704) and the era flags carry almost no signal on their own. Several pairs are strongly correlated (grid and qualifying position 0.97, driver and team standings above 0.9). We keep them: logistic regression is regularised, trees tolerate collinearity, and the ablation in Section 6 tests each group by removal.

Six features show almost no signal on their own (single-feature AUC between 0.49 and 0.56): pit_lane_start, qual_missing, season_opener, grid_minus_qual and the two era flags. A feature can still be useful when it changes how the model reads another feature, so each was kept for a specific hypothesis, which Table 3a checks on the training rows. The evidence supports all of them. Penalised drivers are the clearest case: at the same grid slot they finish on the podium more than twice as often as drivers who qualified there, because a fast car sent down the grid is still a fast car.

Table 3a. *Evidence for the features with no single-feature signal (training rows, 2003 to 2019).*

Feature	Evidence on the training rows, and why the feature is kept
pit_lane_start	20 pit-lane starters, none on the podium (14.2% for everyone else; 0 of 73 since 2003). Marks the last-slot placement as a penalty, not slow pace.
grid_minus_qual	Top-5 qualifiers moved back by a penalty reach the podium 20.3% of the time (64 rows) against 51.0%, yet at the same grid slot they reach it 7.3% of the time against 2.8% for drivers who qualified there. Separates a fast car starting further back from a slow one.
season_opener	At round 1 the standing position is missing and the points share is 0 for 100% of rows (25% otherwise). Tells the model that zero points means the season has not started.
qual_missing	20 training rows have no qualifying record, none on the podium. Lets the model rely on grid and form when qualifying is absent.
era_2003_08, era_2009_13	Podium rate from pole 79.0%, 77.7% and 83.5% in 2003 to 2008, 2009 to 2013 and 2014 to 2019; from third 49.5%, 43.6% and 60.0%. Lets the training fit adjust to each regulation era; the flags are constant on validation and test (Section 6).

4.1 Rejected candidates

Table 4. *Candidates considered and rejected.*

Candidate	Decision and reason
Raw year or season	Rejected: lets the model extrapolate a trend into unseen test years
Raw championship points	Rejected: the scale changed in 2010; share of leader used instead
Home-race flag	Rejected: no detectable effect after grid control (Section 3.2)
Circuit front-row conversion rate	Rejected: circuits are not distinguishable (Section 3.1); a per-circuit estimate from few races would also risk leakage
driverId, constructorId, circuitId as categories	Rejected: memorised identities, hundreds of sparse columns, new drivers every year. Experiment D confirms no gain
Same-weekend sprint results	Rejected: a judgement call. Sprints exist only from 2021, so the column would be empty for almost all training rows, and the sprint-adjusted grid is already reflected in the Sunday grid
Raw qualifying seconds	Rejected: depends on track and weather; percent gap used
Finish position, points, laps, times, fastest lap, this race's status, this race's laps and pit stops	Rejected: post-race outcomes that exist only after the race (positionOrder defines the target)
Standings after this race; rolling form without shift(1)	Rejected: contain the race result (see the leakage experiment below)

4.2 Leakage control

WHAT DATA LEAKAGE IS

Data leakage is the use of information that would not exist at prediction time. It inflates notebook scores and fails in real use. The classic case is a rolling “last 5 races” average whose window includes the current race, and therefore the answer.

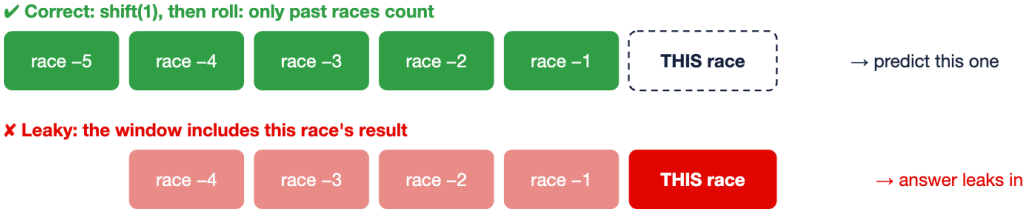


Diagram 3. *shift(1) before rolling: the window contains only earlier races, never the race being predicted.*

Three safeguards back the cutoff argument. First, every rolling and expanding statistic is shifted by one race, and the standings merge is asserted to use an earlier round. Second, a future-invariance test rebuilds features for 8 target races after deleting all later races and scrambling that race's outcomes and post-race standings: none of the 28 features changed in any of the 8 races, while the same test does detect a deliberately unshifted rolling rate (control). Third, Experiment E measures what leakage would buy: adding an unshifted rolling podium rate lifts validation ROC-AUC from 0.933 to 0.997, and post-race standings lift it to 0.973. Those values are unrealistically high, which is why the safe pipeline scores lower and why we trust it.

4.3 Preprocessing and preprocessing-order experiments

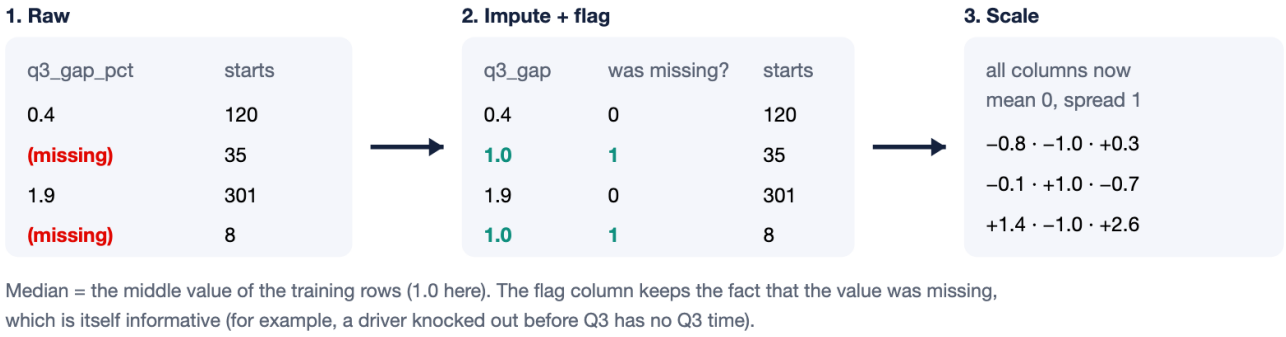


Diagram 4. *Preprocessing: median imputation with a missing-value flag, then standard scaling. Both are fitted on the training rows only.*

The preprocessing fits only on training rows: median imputation with a missing-value flag for each of the 17 features that have gaps in training, followed by standard scaling, which turns the 28 features into 45 model columns. Missing values are common (for example q2_gap_pct is empty in 37% and q3_gap_pct in 62% of training rows, because drivers eliminated earlier have no later session and some seasons used other formats), so the flags let a model use missingness as information. Median imputation was chosen because it is robust to the long tails visible in Table 5, and Experiment B compares it with zero fill.

Table 5. *Effect of median imputation and standard scaling on nine representative features, computed on the 6,783 training rows (2003 to 2019) on which the imputer and scaler are fitted.*

Feature	Missing (train)	Raw mean (sd)	Raw range	Median fill	Range after scaling (sd units)
grid_clean	0.0%	11.18 (6.23)	1 to 24	11	-1.6 to +2.1
qual_pos	0.3%	11.15 (6.22)	1 to 24	11	-1.6 to +2.1
q1_gap_pct	1.8%	1.78 (1.82)	0 to 33.3	1.34	-1.0 to +17.4
q2_gap_pct	37.4%	1.31 (1.16)	0 to 16.1	1.12	-1.3 to +16.2
q3_gap_pct	61.9%	1.18 (1.20)	0 to 33.0	1.01	-1.4 to +43.0
drv_podium_rate_20	3.3%	0.145 (0.225)	0 to 1	0.00	-0.6 to +3.9
drv_starts_prior	0.0%	92.9 (79.2)	0 to 321	70	-1.2 to +2.9
drv_standing_pos_before	7.1%	11.27 (6.46)	1 to 27	11	-1.6 to +2.5
con_mech_rate_20	0.4%	0.118 (0.093)	0 to 0.75	0.10	-1.3 to +6.8

The step changes values in three ways (Figure 10a). Imputation replaces each gap with the training median: 37.4% of training rows end up on a single value for q2_gap_pct and 62.0% for q3_gap_pct, which shrinks their standard deviations from 1.16 to 0.92 and from 1.20 to 0.74, and the missing-value flag keeps the information that the value was filled. Scaling puts columns with ranges as different as 0 to 1 and 0 to 321 on the same unit scale, which matters for the logistic regression and the network and not for the forest. Scaling does not remove tails: a Q3 gap of 33% stays 43 standard deviations above the mean and a Q1 gap of 33% stays 17, a reminder that rare extreme rows remain in the data.

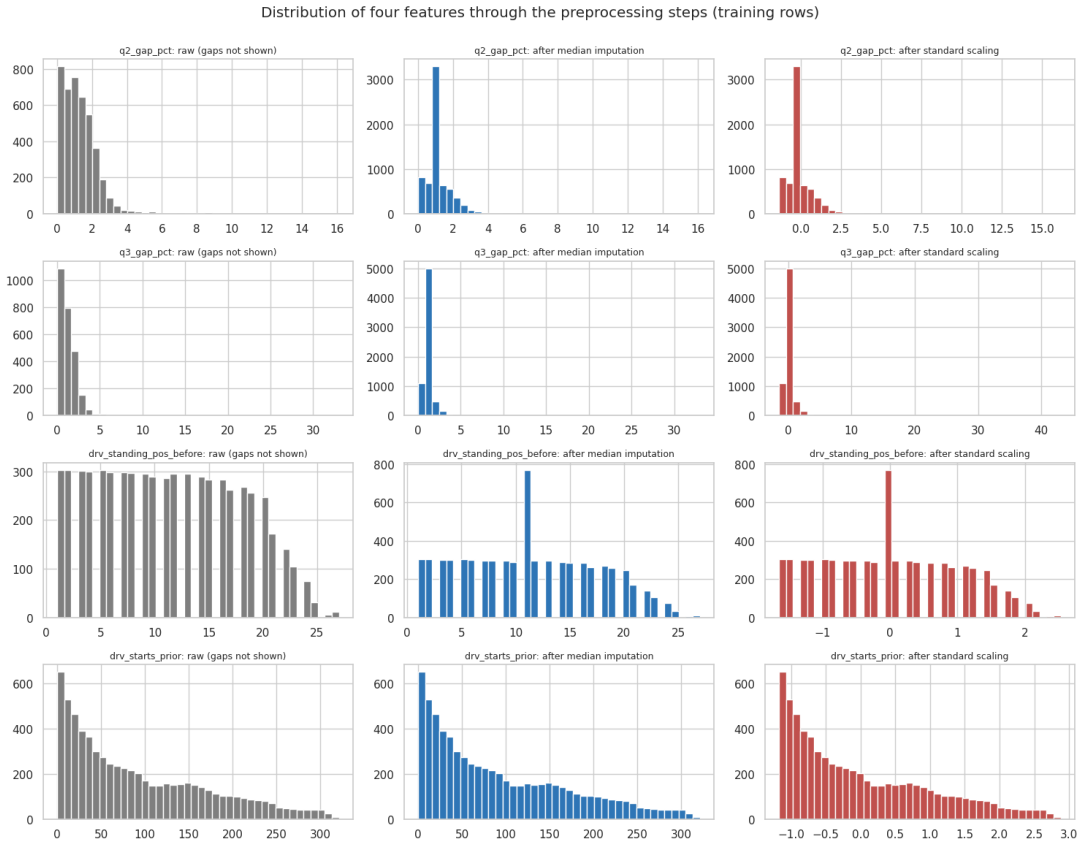


Figure 10a. *Distribution of q2_gap_pct, q3_gap_pct, drv_standing_pos_before and drv_starts_prior on the training rows: raw (gaps not shown), after median imputation and after standard scaling.*

WHAT THE CHART SHOWS

Each row follows one feature through the preprocessing: raw values (missing values not shown), after median imputation and after standard scaling. Imputation creates a tall spike at the median wherever many values are missing, as for the Q2 and Q3 gaps. Scaling changes only the units on the horizontal axis; the shape of the distribution and its long tail are unchanged.

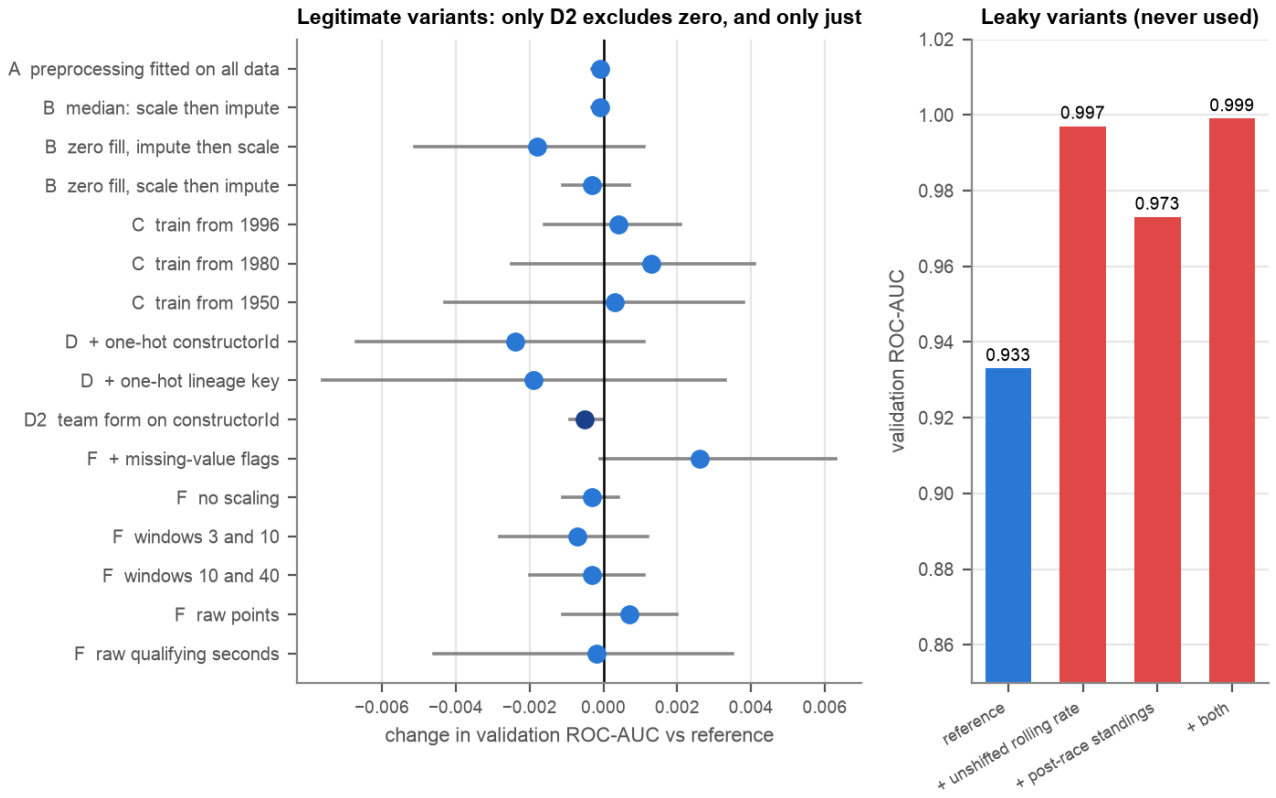


Figure 10. *Left: effect of each preprocessing, training-window or encoding variant (Experiments A to D2 and F) on validation ROC-AUC, with 95% intervals from resampling races. Every interval includes zero except D2, whose interval only just excludes it (-0.0009 to -0.0000). Right: the same measure for three leaky variants, which are never used.*

WHAT THE CHART SHOWS

Each point shows how much one variant changes validation ROC-AUC compared with the reference pipeline, with a 95% interval from resampling races. Every interval in the left panel crosses zero except D2, which only just excludes it, so none of the preprocessing, window or encoding choices makes a meaningful difference. The right panel shows the leaky variants, which reach 0.973 to 0.999 against an honest 0.933; a jump of this size is the typical sign of leakage.

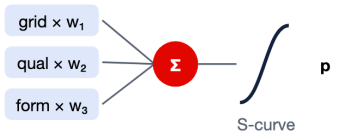
Fitting the imputer and scaler on all data instead of training rows only changes validation ROC-AUC by -0.0001. Imputing before or after scaling gives the same result for median imputation by algebra (-0.0001), and for zero fill the order costs -0.0018 (impute then scale) or -0.0003 (scale then impute, equal to mean imputation). Adding team identity as one-hot columns on top of the form features (Experiment D) adds 31 columns for constructorId, leaves 21.3% of validation rows with a team never seen in training (0.0% with the lineage key) and scores no better than the reference (-0.0024 for constructorId, -0.0019 for the lineage key). Experiment D2 tests the lineage key itself: rebuilding the four team form features on the raw constructorId changes them for 4.4% of training rows and lowers validation ROC-AUC by 0.0005 (95% CI -0.0009 to -0.0000). The loss is small but consistent across resamples, so the lineage key is kept. The preprocessing order therefore matters little here, but we keep the strict rule of fitting on training rows because it is the correct protocol and costs nothing.

Experiment F tests the remaining design choices that had only been reasoned, each with one retraining of the reference logistic regression on validation. Adding missing-value flags changes validation ROC-AUC by +0.0026 (95% CI -0.0001 to +0.0063), dropping scaling by -0.0003, rolling windows of 3 and 10 races by -0.0007, windows of 10 and 40 by -0.0003, raw championship points by +0.0007 and raw qualifying seconds by -0.0002. All intervals include zero. None of these choices moves validation performance beyond noise, so each is kept for its stated reason (comparability across eras and tracks, and scaled inputs for the logistic regression and the network). The missing-value flags show the largest, though not reliable, gain, which supports keeping them in the modelling pipeline.

5 Predictive modelling

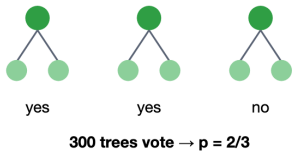
5.1 Models, baseline and metric choice

All three models are trained on the same modeling table: the same 6,783 training rows, the same 775 validation rows and the same 28 features (45 columns after preprocessing). The random forest was chosen as the second statistical model because it captures non-linear effects and interactions without feature engineering and offers exact tree SHAP values. The test seasons were not used for any decision: hyperparameters were chosen on validation PR-AUC, thresholds were chosen on validation, and the test set was scored once at the end.



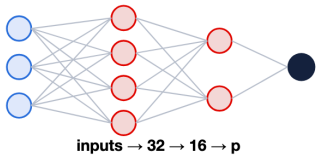
Logistic regression

Weighted sum of the inputs passed through an S-shaped curve. The baseline.



Random forest

300 decision trees on random samples of rows and features; their votes are averaged.



Shallow neural network

Two hidden layers (32 and 16 units) learned by gradient descent with dropout and early stopping.

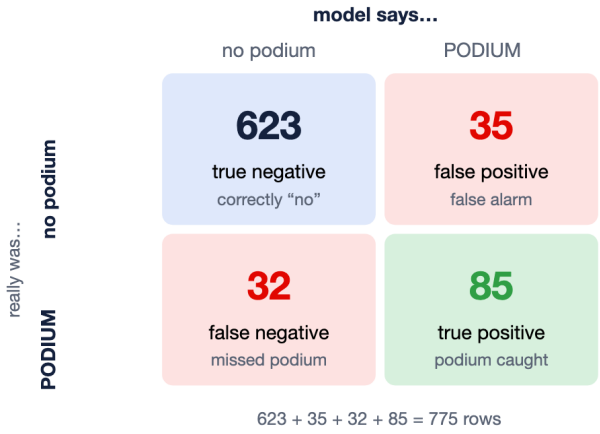
Table 6. *The three model families.*

Model	How it works	Main limitation here
Logistic regression	Weighted sum of the inputs passed through a sigmoid; coefficients are fitted by regularised maximum likelihood. Used as the baseline (A0, C = 1)	Linear in log-odds, so interactions such as grid by form must be learned indirectly; median filling makes it swing with missing inputs
Random forest	Many decision trees on random samples and feature subsets, averaged. Tuned to depth 8 and at least 50 rows per leaf	Probabilities are coarse; without depth and leaf limits it memorises the training seasons (train PR-AUC 1.00)
Shallow neural network	Dense layers of 32 and 16 ReLU units with dropout 0.3, L2 1e-4, early stopping and a positive class weight of 6 (attempt A8)	Seed noise of about 0.011 PR-AUC; class weighting makes its probabilities uncalibrated, so it is best read as a ranking score

The baseline reference is a simple rule, podium if the grid slot is 1 to 3, and attempt A0, a logistic regression with default settings on all features. Podiums are rare (14.2% of training rows, 15.1% in validation and test), so PR-AUC is the primary metric: it measures how well the model ranks true podium finishers above the rest and is far more sensitive to errors in the rare class than ROC-AUC. ROC-AUC is secondary and describes overall separation. F1, precision and recall are reported at one threshold per model, chosen on validation only as the value from 0.02 to 0.90 that maximises F1. Because the drivers of one race are not independent (three podium places are shared among them), all confidence intervals (300 resamples) resample whole races, not rows.

How the metrics work

The models output a podium probability for every driver. A threshold turns it into a yes or no; the threshold-free metrics judge the ranking of the scores themselves. The confusion matrix below uses the selected random forest on the 775 validation rows at its threshold of 0.36.



Precision = 85 / (85 + 35) = 0.71

When the model says podium, how often is it right? 71% of its podium calls were correct.

Recall = 85 / (85 + 32) = 0.73

Of all real podiums, how many did it catch? It caught 73%.

F1 = 0.72

The harmonic mean of precision and recall; high only when both are high. The threshold maximises F1 on validation.



Diagram 5. *The threshold trade-off: a lower threshold catches more podiums (recall) at the cost of more false alarms (precision).*

ROC-AUC

The probability that a randomly chosen podium finisher receives a higher score than a randomly chosen non-podium driver. 0.5 is chance and 1.0 is perfect. It is flattered by the many easy non-podium rows.

PR-AUC (primary)

Average precision across all recall levels. It concentrates on the rare positive class; its no-skill level equals the podium rate (about 0.15), not 0.5. This is why it is the primary metric.

Why not accuracy?

About 85% of rows are not podiums, so a model that always answers “no podium” is 85% accurate and finds no podium at all.

5.2 Modelling attempts

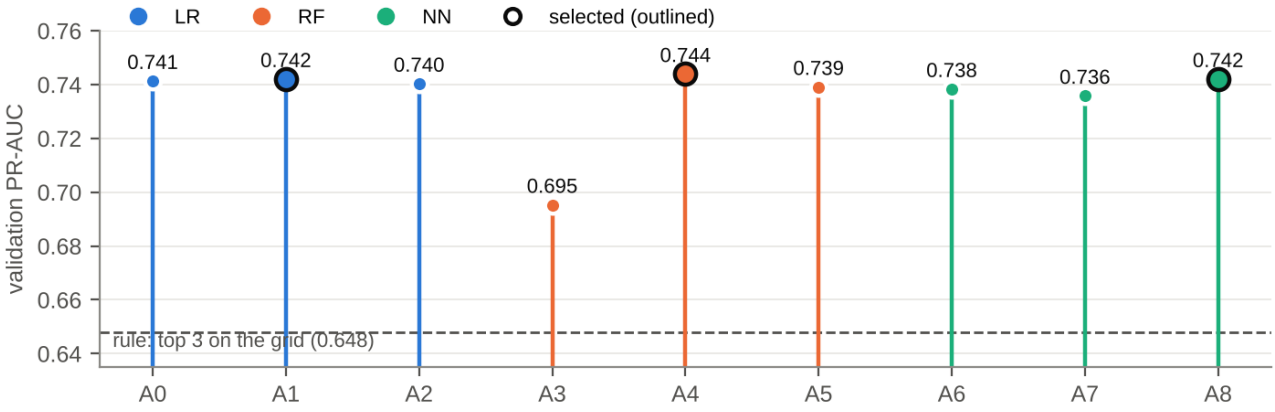


Figure 11. *Validation PR-AUC for nine attempts against the grid rule. A0 to A2: logistic regression (baseline, tuned C, balanced weights). A3 to A5: random forest (unlimited depth, regularised, balanced). A6 to A8: neural network (1x16 plain, 32-16 with dropout, with class weight). The three outlined attempts were selected per family.*

WHAT THE CHART SHOWS

Each stem shows the validation PR-AUC of one attempt, coloured by model family, and the dashed line marks the grid rule at 0.648. Every attempt scores above the rule. The only clear drop is A3, the unrestricted forest that memorised the training data; limiting tree depth and leaf size (A4) recovers it. The outlined attempts are the ones selected in each family.

Table 7. *Before and after for every attempt on the validation seasons. Intervals resample races; an interval that includes zero means no reliable change.*

Attempt	What changed	Val PR-AUC	Val ROC	Val F1	Compared with	Change in PR-AUC (95% CI)	Reading
A0	LR, C = 1, all features (baseline)	0.741	0.936	0.714			
A1	LR, C = 0.3, tuned on validation	0.742	0.936	0.709	A0	+0.0007 (-0.0013 to +0.0035)	Flat plateau for C from 0.1 to 10; no gain
A2	LR with balanced class weights	0.740	0.935	0.711	A1	-0.0016 (-0.0092 to +0.0051)	Only moves the threshold (0.24 to 0.64)
A3	RF, 300 trees, unlimited depth	0.695	0.928	0.707	A0	-0.0459 (-0.0909 to -0.0023)	Hurt: memorises training (train PR-AUC 1.00)
A4	RF, depth 8, leaf 50, sqrt features	0.744	0.935	0.717	A3	+0.0486 (+0.0115 to +0.0766)	Helped: limits close the overfitting gap
A5	A4 with balanced subsample weights	0.739	0.933	0.721	A4	-0.0049 (-0.0149 to +0.0066)	No gain; threshold shifts to 0.77
A6	NN, 1x16 ReLU, 60 epochs	0.738	0.930	0.716	A0	-0.0030 (-0.0456 to +0.0327)	Matches the baseline
A7	NN, 32-16, dropout 0.3, L2, early stop	0.736	0.929	0.704	A6	-0.0025 (-0.0317 to +0.0198)	Closes the train gap, no accuracy gain
A8	A7 plus positive class weight 6	0.742	0.930	0.702	A7	+0.0062 (-0.0146 to +0.0287)	Rebalances precision and recall; no PR gain

The logistic regression was tuned over C from 0.003 to 10 (validation PR-AUC rises from 0.721 to a plateau near 0.742 from C = 0.1) and the forest over 12 combinations of depth (4, 6, 8, 12) and minimum leaf size (5, 20, 50); larger leaves consistently scored better, so the best setting (depth 8, leaf 50) is a regularisation choice. The networks stay within the brief's one or two hidden layers and use Adam (learning rate 0.001, batch size 128), because 6,783 training rows cannot support a deep model.

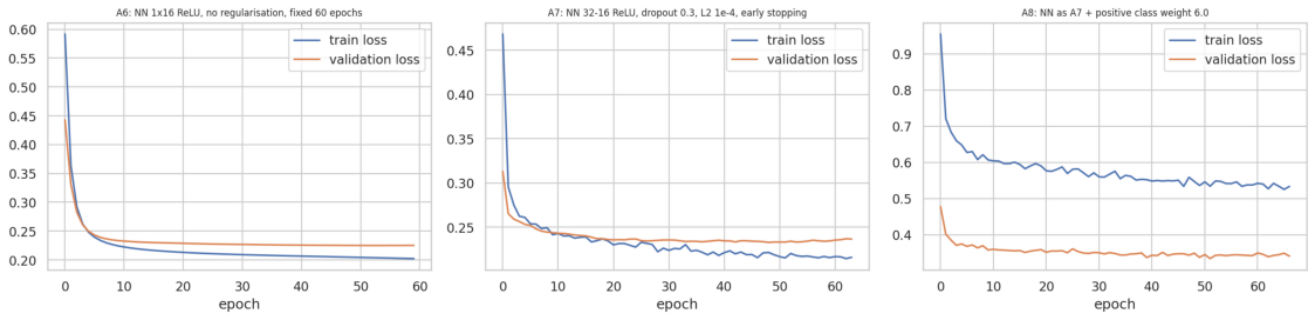


Figure 12. *Training and validation loss per epoch for the three neural-network attempts (A6 plain, A7 with dropout, L2 and early stopping, A8 with class weight).*

WHAT THE CHART SHOWS

Each panel shows the training and validation loss per epoch for one network attempt. In all three, both curves fall and then level off without the validation loss rising again, so the networks do not overfit strongly; A7 and A8 stop early at their best epoch. In A8 the training loss stays above the validation loss because the positive class weight of 6 inflates the training loss, not because the network underfits.

Hyperparameters were chosen on validation PR-AUC only. Three results stand out. Tuning the logistic regression's C from 1 to 0.3 changes nothing reliably (+0.0007, CI -0.0013 to +0.0035). The unrestricted random forest overfits (train PR-AUC 1.000, validation 0.695, gap 0.305), and the regularised forest recovers +0.049 (CI +0.012 to +0.077), the one clearly helpful change. All neural-network variants are within seed noise of each other: five seeds of one configuration give validation PR-AUC 0.7317 plus or minus 0.0111. Every attempt is above the rule (0.648), and the spread among the selected three is 0.002. The learning curves show no strong overfitting: the plain network (A6) reached its lowest validation loss at epoch 52 of 60 and ended at the same value, and the regularised network (A7) stopped after 64 epochs with its best epoch at 49.

Because most of these differences are within noise, the selected attempt in each family was also tested against every other attempt in its family with the same race-level bootstrap. Only one comparison is reliable: the regularised forest A4 beats the unrestricted A3 (+0.049, CI +0.012 to +0.077). A1 is tied with A0 and A2, A4 is tied with A5, and A8 is tied with both A6 (+0.0037, CI -0.018 to +0.026) and A7. Where attempts are tied, the selection rests on design rather than score: A1 is the baseline model with its penalty chosen by search instead of the library default, A4 is the regularised forest, and A8 stops at the epoch with the lowest validation loss and uses dropout and L2, instead of training for an arbitrary fixed 60 epochs. Every selected model is compared with the baseline A0 on the test seasons.

5.3 Results on training, validation and test

Table 8. *Training and validation (2020 to 2021) scores for all five metrics. Thresholds were chosen on validation.*

Model	Train ROC	Train PR	Train F1	Val ROC	Val PR	Val F1	Val prec.	Val rec.	Thr.
LR A0 (baseline)	0.934	0.699	0.673	0.936	0.741	0.714	0.659	0.778	0.24
LR A1 (selected)	0.934	0.698	0.673	0.936	0.742	0.709	0.657	0.769	0.24
RF A4 (selected)	0.944	0.728	0.693	0.935	0.744	0.717	0.708	0.727	0.36
NN A8 (selected)	0.945	0.723	0.673	0.930	0.742	0.702	0.721	0.684	0.82
Rule: top 3 on grid	0.909	0.628	0.648	0.910	0.648	0.695	0.698	0.692	-

Table 9. *Unseen test seasons (2022 onward), scored once with validation thresholds.*

Model	Test ROC-AUC (95% CI)	Test PR-AUC (95% CI)	F1	Precision	Recall
LR A0 (baseline)	0.926 (0.904 to 0.946)	0.712 (0.662 to 0.770)	0.680	0.565	0.853
LR A1 (selected)	0.926 (0.903 to 0.945)	0.711 (0.662 to 0.770)	0.675	0.560	0.848
RF A4 (selected)	0.938 (0.920 to 0.954)	0.720 (0.666 to 0.780)	0.683	0.627	0.750
NN A8 (selected)	0.928 (0.909 to 0.945)	0.698 (0.651 to 0.756)	0.650	0.617	0.686
Rule: top 3 on grid	0.913	0.647	0.649	0.650	0.647

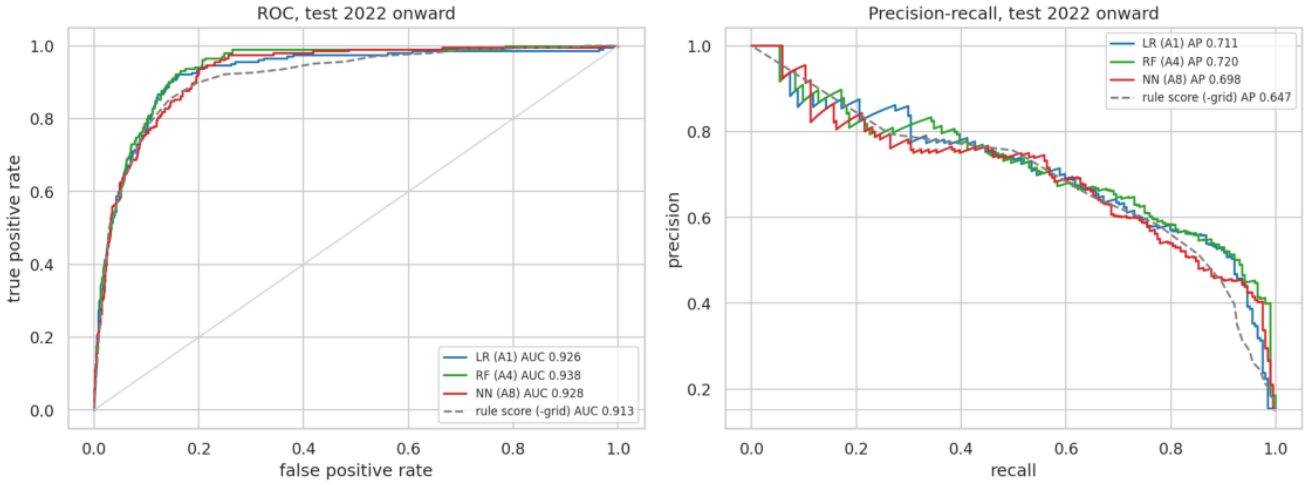


Figure 13. *ROC (left) and precision-recall (right) curves on the test seasons for the three selected models and the grid rule.*

WHAT THE CHART SHOWS

The ROC curves (left) show, for every threshold, the share of podiums found against the share of false alarms; the precision-recall curves (right) show precision against recall. The three model curves lie close together and slightly above the dashed grid rule over most of the range. The chart therefore supports two conclusions: the models improve modestly on the rule, and they cannot be separated from each other.

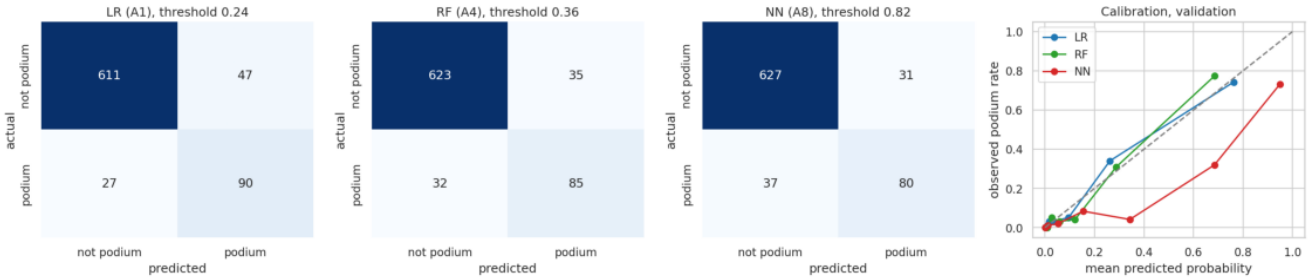


Figure 14. *Validation confusion matrices at each model's threshold (LR, RF, NN) and calibration curves. The NN curve lies below the diagonal, so its probabilities overstate the podium chance.*

WHAT THE CHART SHOWS

The confusion matrices count correct and incorrect calls on the validation seasons at each model's threshold, with correct calls on the diagonal. The forest has the most balanced errors, while the logistic regression finds more podiums at the cost of more false alarms. In the calibration panel a well-calibrated model lies on the dashed diagonal; the network's curve lies below it, so its probabilities overstate the podium chance and are best used as a ranking score.

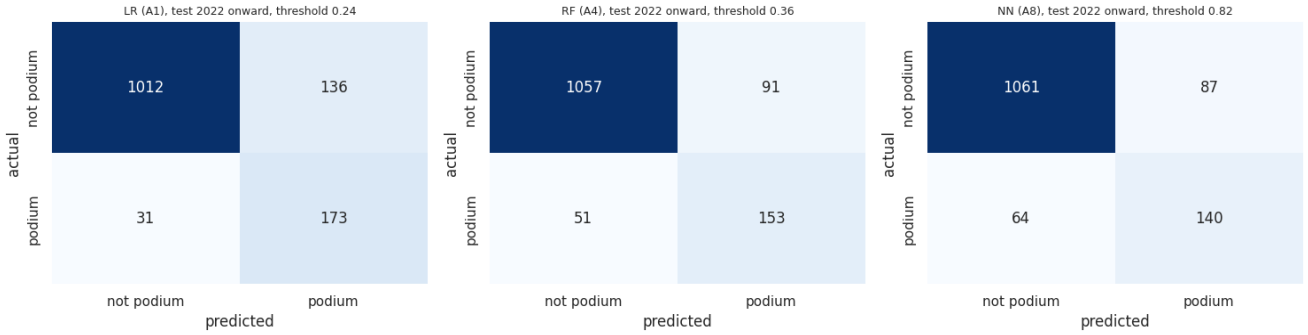


Figure 14a. *Test confusion matrices (2022 onward) for the three selected models at their validation thresholds.*

WHAT THE CHART SHOWS

On the 1,352 test rows, which include 204 podiums, the forest finds 153 podiums with 91 false alarms. The logistic regression finds more podiums (173) but raises 136 false alarms, and the network finds the fewest (140) with 87 false alarms. Each model keeps the balance of errors it showed on validation, with somewhat lower precision, which matches the drop from validation to test F1 discussed below.

On test, all three models beat the rule on ROC-AUC (0.926 to 0.938 against 0.913) and on PR-AUC (0.698 to 0.720 against 0.647), but we ran no paired test against the rule and the model intervals are about 0.1 wide, so the PR-AUC gain is modest and not proven; the intervals also overlap each other. Paired comparisons against the baseline give one reliable difference: the random forest's ROC-AUC is +0.0126 higher (CI +0.0023 to +0.0282). The remaining paired PR-AUC and ROC-AUC differences all include zero. So the honest summary is that learned models add a modest improvement over a strong rule, and that choosing among the three rests on secondary arguments (stability, interpretability) and not on accuracy.

We nominate the random forest as the primary model, with the other two as comparators. It has the best validation PR-AUC (0.744), the best test ROC-AUC and PR-AUC, the only reliable gain over the baseline, exact SHAP explanations, and with reduced models for unknown inputs it is the only family whose grid-only prediction for a pole sitter matches the historical rate (0.803 against 80.3%, Section 8). The nomination is a judgement based on consistency across these checks and not on a statistically proven accuracy gap.

Two facts help in reading Table 9. Test F1 (0.65 to 0.68) sits below validation F1 (0.70 to 0.72) because thresholds were tuned on the small validation set; this can move either way without a fault. The test seasons also follow regulations the training data never saw, which is a real distribution shift. The validation score of the neural network is slightly optimistic because early stopping also used validation.

6 Feature-group ablation

View 1 · Drop one group
Train with every group except one. A large drop means the model needed that group, unless another group carries the same information.

View 2 · Only one group
Train with a single group to measure what it can do on its own.

View 3 · Build-up
Add groups in a fixed order (grid, qualifying, form…) and record the gain from each addition.

Each group was removed in turn and the model retrained (three seeds), and the groups were also added one by one and tested alone. The measure is the change in validation PR-AUC, with 95% intervals from resampling races. Filled markers mean the interval excludes zero.

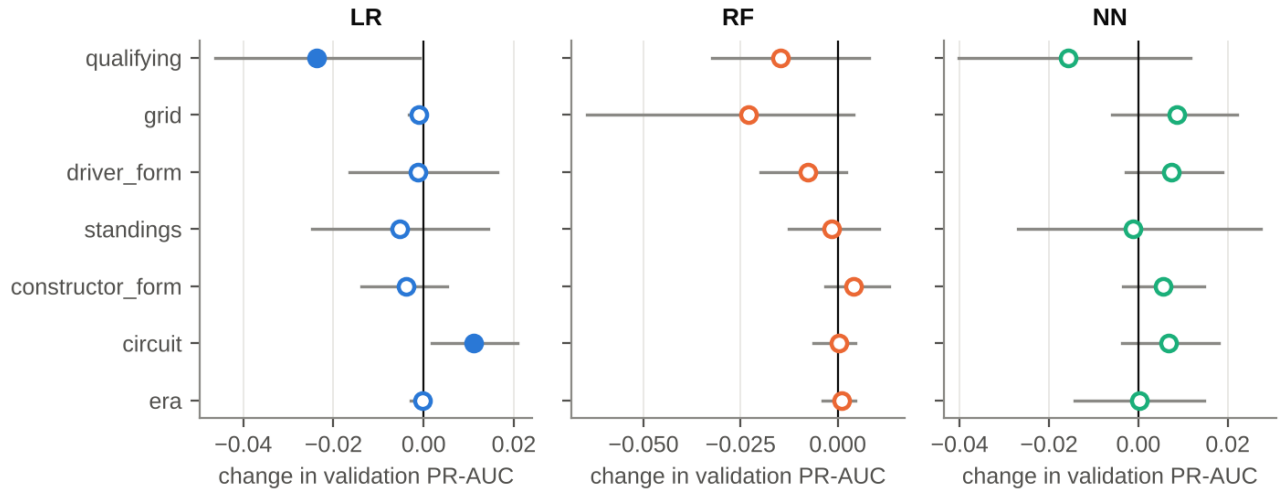


Figure 15. *Change in validation PR-AUC when one feature group is removed (negative means the group helped). Full-model PR-AUC: LR 0.742, RF 0.742, NN 0.737. Each panel uses its own horizontal scale.*

WHAT THE CHART SHOWS

Each point shows the change in validation PR-AUC when one feature group is removed and the model is retrained; a value left of zero means the group was useful. Filled points have intervals that exclude zero. Only two changes are reliable, both for the logistic regression: removing qualifying lowers PR-AUC by 0.024, and removing the circuit group raises it by 0.011.

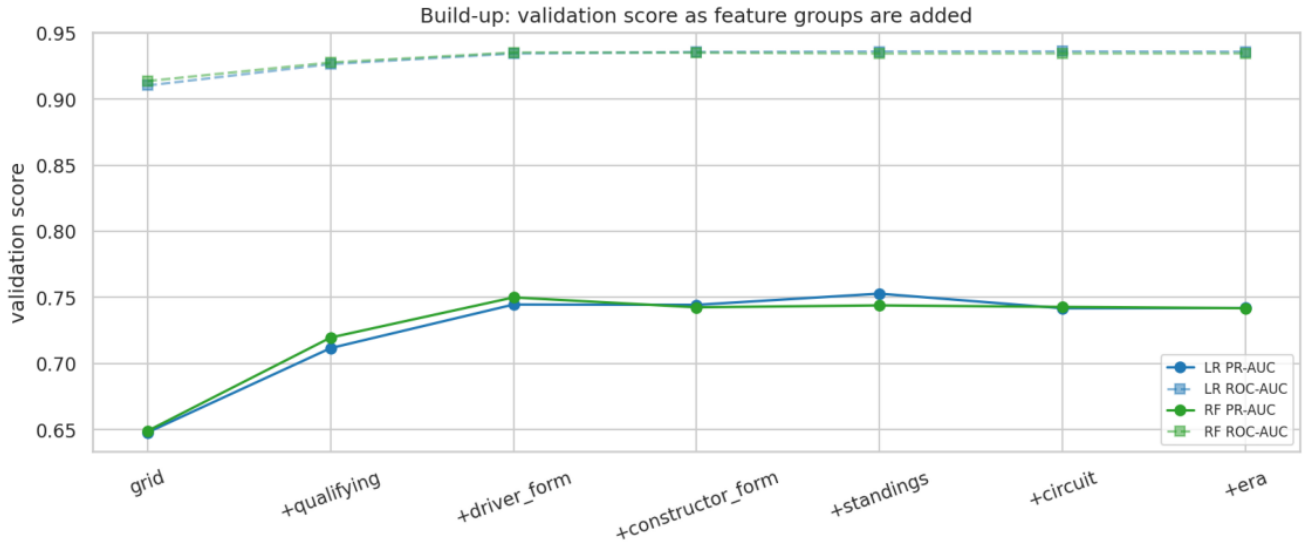


Figure 16. *Build-up: validation PR-AUC (solid) and ROC-AUC (dashed) as groups are added in order, for LR and RF.*

WHAT THE CHART SHOWS

The solid lines show validation PR-AUC and the dashed lines ROC-AUC as feature groups are added from left to right. The largest step is adding qualifying to the grid (+0.064 for LR and +0.071 for RF), followed by driver form. After driver form the lines are essentially flat, so the remaining groups add little once the strong groups are present.

Qualifying is the most valuable group: removing it costs the logistic regression 0.024 PR-AUC (the only reliable drop), and adding it to the grid raises PR-AUC by +0.064 (LR) and +0.071 (RF), both reliable. Driver form comes second (adding it gives +0.033 for LR and +0.030 for RF, the latter reliable). The grid group matters most for the random forest when removed (-0.023, not reliable) but is largely replaced by qualifying position in the linear models, since the two correlate at 0.97. Constructor form and standings add nothing reliable once the other groups are present. The era result is empty by construction: both era flags are 0 for every validation and test row (all from 2014 on), so neither the ablation nor permutation importance can measure them.

The circuit group is the surprise: removing it improves the logistic regression by +0.011 (CI +0.002 to +0.021), and adding it as the sixth step lowers PR-AUC by 0.011. The circuit history is noisy and sparse (24% to 28% missing in train and validation). The gain appears only for the logistic regression; for the forest (+0.0004) and the network (+0.0067) the change is within noise. The delivered models keep the group because they were selected on validation in Section 5 before this ablation ran and before the test set was scored; dropping the group now would mean choosing a model after seeing test results. A model without the circuit group is the natural next step. Two cautions apply to the whole section: groups overlap, so a small drop does not prove a group is worthless, and validation has only about 40 races, so most intervals are wide and the ordering is indicative.

7 Explainability

EXPLANATIONS ARE NOT CAUSAL EFFECTS

Every explanation below describes what the trained models rely on to predict. A feature being important to a model is not proof that it causes podium finishes. A good grid slot comes from a fast car, a fast lap and a strong team, and the models cannot separate the driver from the car. Qualifying position and grid position are nearly the same variable (correlation 0.97), so how a model splits credit between them is arbitrary. Nothing here supports a claim such as 'qualifying better causes more podiums'.

Permutation importance

Shuffle one feature (or group) among the validation rows and measure the fall in PR-AUC. A large fall means the model leans on it.

SHAP

Splits each prediction into an additive push from every feature; the pushes plus the base value equal the model output exactly.

LIME

Perturbs one row thousands of times and fits a small linear model around it; a local second opinion.

Permutation importance uses 10 shuffles per feature or group. SHAP is exact for the logistic regression and the forest on all 775 validation rows, and approximate for the network (Kernel SHAP on 150 rows, with a 15-cluster summary of the training data as background). LIME uses 3,000 perturbed samples per case. All explanations use the 2020 to 2021 validation rows, so the test seasons stay unseen, and a different period could rank features differently.

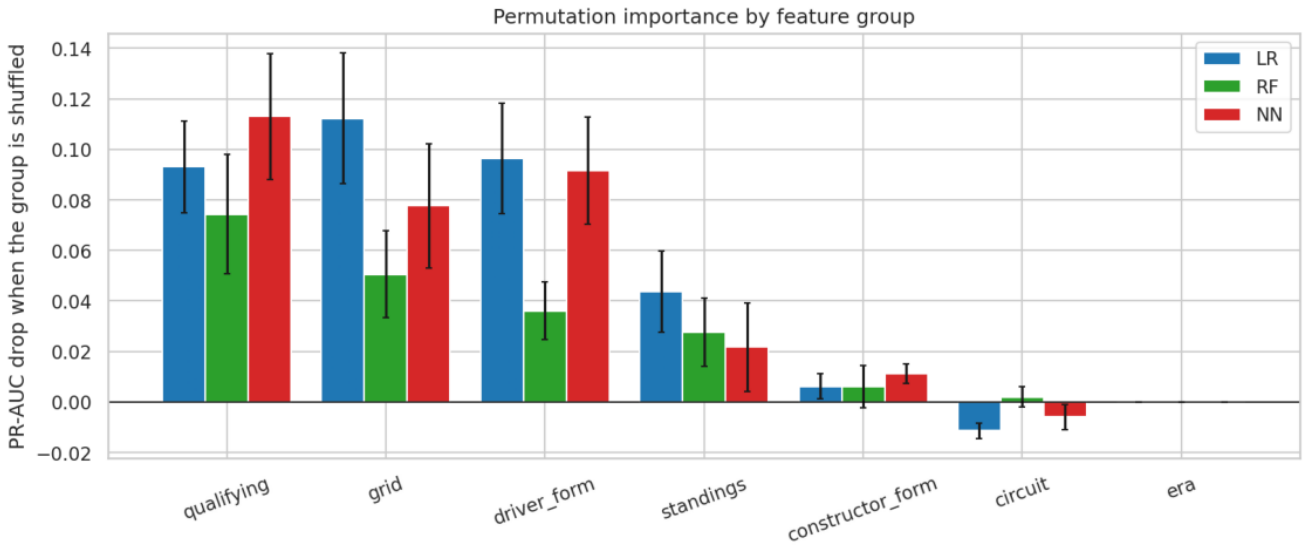


Figure 17. *Permutation importance by feature group for the three models (error bars show the spread over repeated shuffles).*

WHAT THE CHART SHOWS

Each bar shows how much validation PR-AUC falls when a whole feature group is shuffled, for each model. Qualifying, grid and driver form dominate in all three models, constructor form and circuit are close to zero, and era is exactly zero because the era flags are constant on the validation seasons.

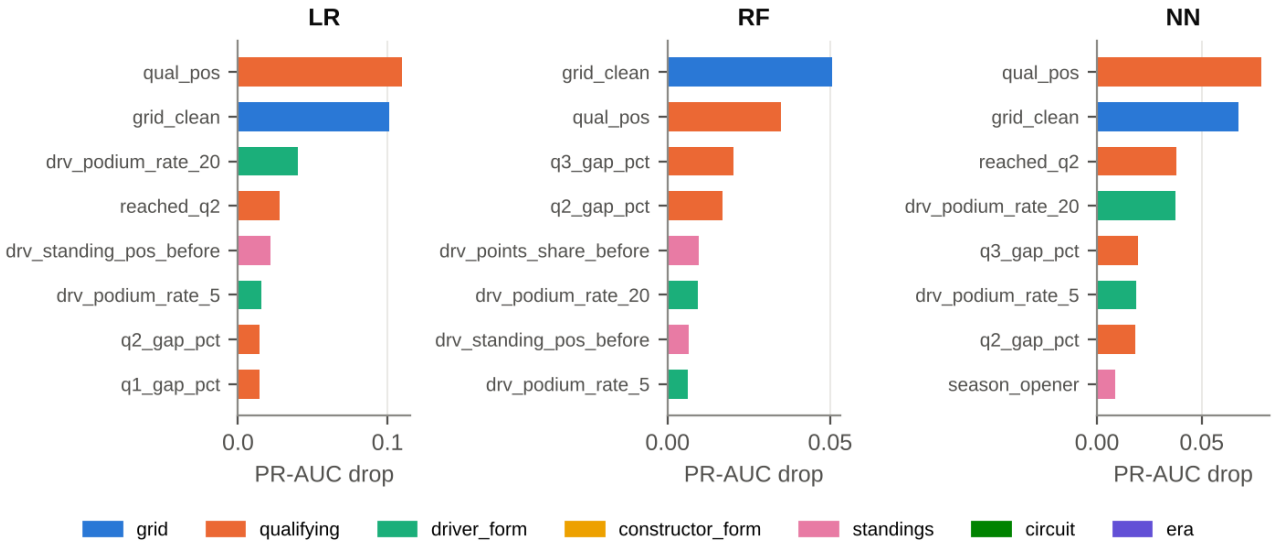


Figure 18. *Top eight features by permutation importance for each model, coloured by group.*

WHAT THE CHART SHOWS

The chart lists the eight features whose shuffling hurts each model most, coloured by group. Grid position and qualifying position lead in every model. Because the two correlate at 0.97, the way a model divides credit between them is arbitrary and should not be over-interpreted.

All three models rely on the same few inputs. Grid position and qualifying position lead in every model (for the forest, grid_clean 0.050 and qual_pos 0.035; for the logistic regression qual_pos 0.110 and grid_clean 0.101), followed by qualifying gaps and recent driver form (drv_podium_rate_20 0.040 for LR). By group, qualifying (0.074 to 0.113), grid (0.050 to 0.112) and driver form (0.036 to 0.096) dominate, standings are smaller (0.022 to 0.044), constructor form is close to zero, circuit is zero or slightly negative (shuffling it did not hurt) and era is exactly zero, because both era flags are constant on the validation rows. Permutation importance and SHAP agree (Spearman 0.66 for LR, 0.81 for RF and 0.67 for NN), and the SHAP rankings of different models correlate between 0.55 and 0.69.

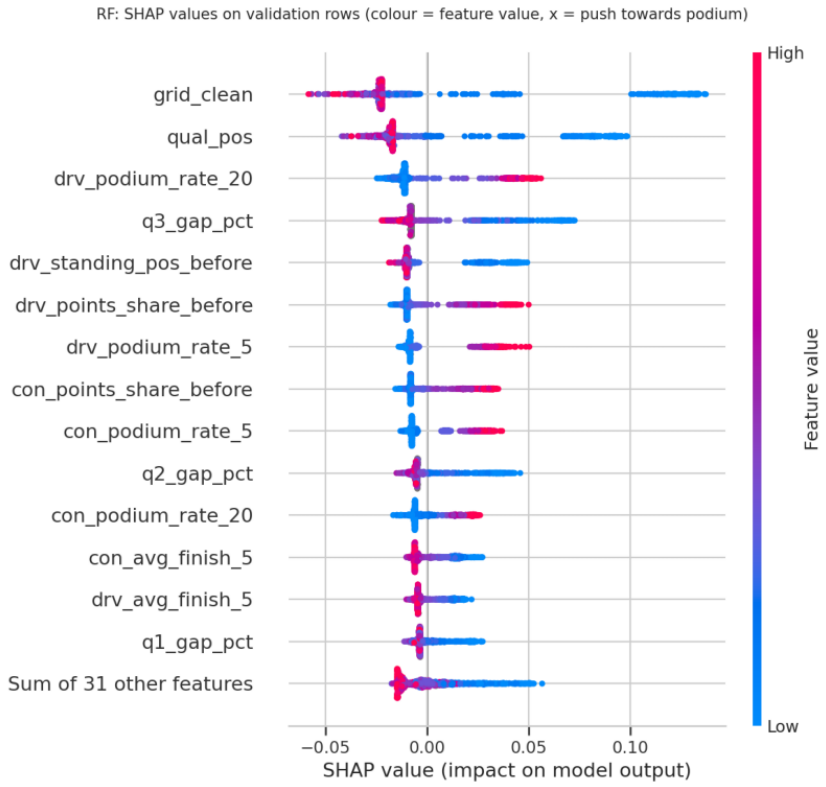


Figure 19. *SHAP values of the random forest on validation rows. Each dot is one driver-race, colour shows the feature value and the horizontal position shows the push towards or away from a podium.*

WHAT THE CHART SHOWS

Each row is a feature, ordered by importance, and each point is one validation driver-race. The horizontal position is the SHAP value, the push towards (right) or away from (left) a podium, and the colour gives the feature's value. For grid_clean the low values (front of the grid, blue) lie far to the right, so starting near the front pushes the prediction strongly towards a podium.

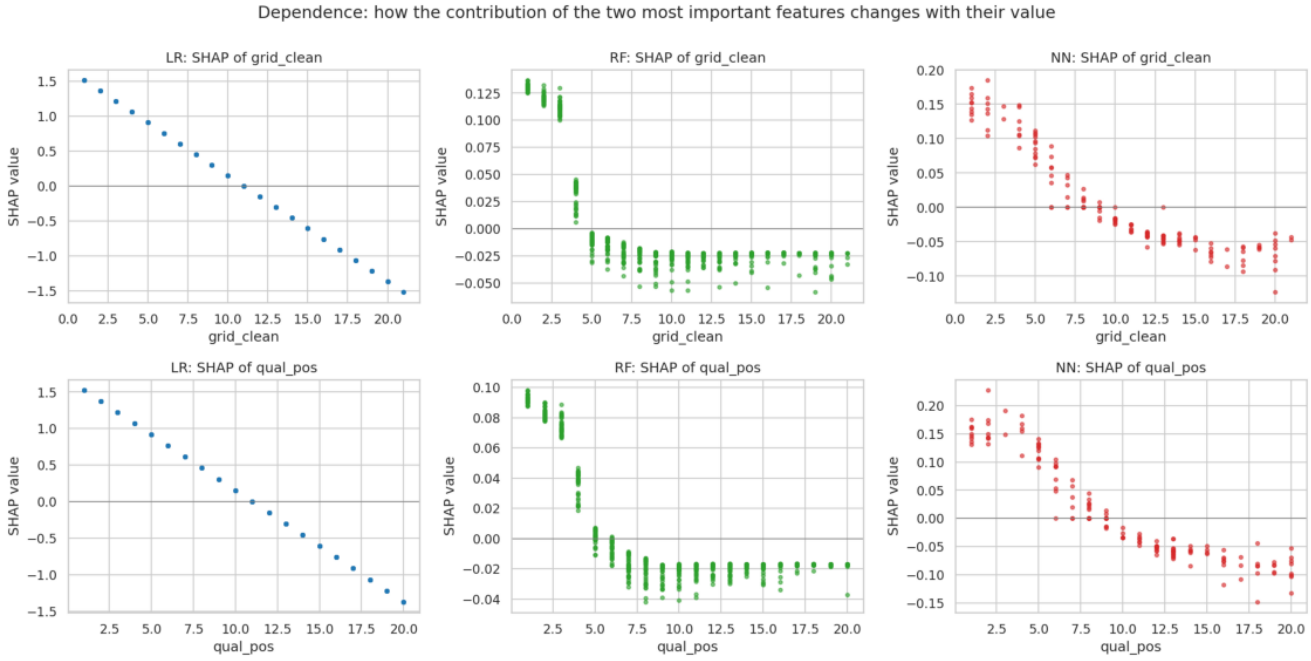


Figure 20. *SHAP dependence of the two top features: contribution against grid_clean (top) and qual_pos (bottom) for LR, RF and NN.*

WHAT THE CHART SHOWS

Each panel plots a feature's value against its SHAP contribution for one model, with grid_clean in the top row and qual_pos in the bottom row. The logistic regression gives a straight line in log-odds, whereas the forest and the network drop sharply up to about fifth place and then level off, which matches the podium rate by slot in Figure 8.

The direction is the same everywhere and sensible: higher grid or qualifying position (a worse slot) lowers the predicted podium chance, and a higher recent driver podium rate or team points share raises it. The dependence plots show the shapes differ by model: the logistic regression is a straight line in log-odds, while the forest and the network step down sharply after about fifth place and flatten beyond it, which matches the podium rate by slot in Figure 8.

7.1 Local explanations

Three validation cases illustrate single predictions with SHAP waterfalls (the forest's base value is 0.142, the average podium rate) and LIME.

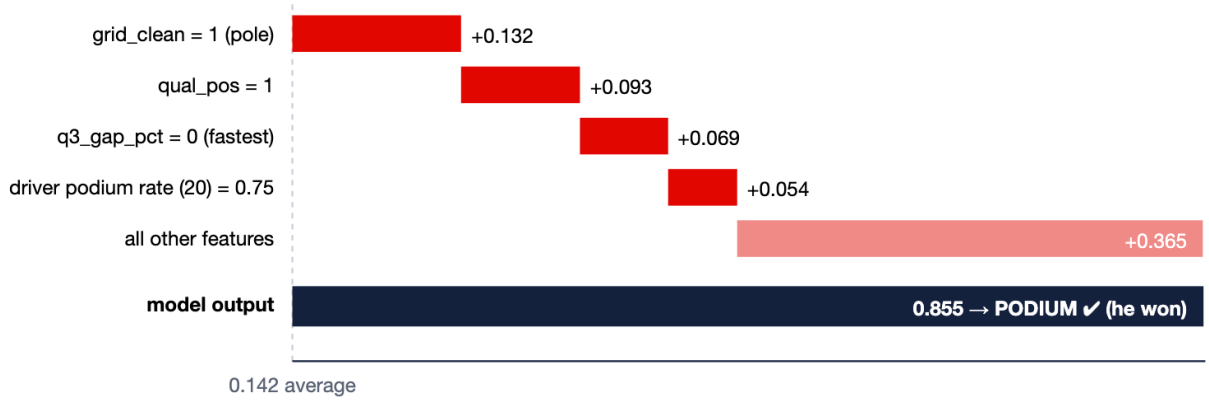


Diagram 6. *Simplified SHAP waterfall for Lewis Hamilton, 2020 Belgian GP: from the average prediction (0.142) to the model output (0.855).*

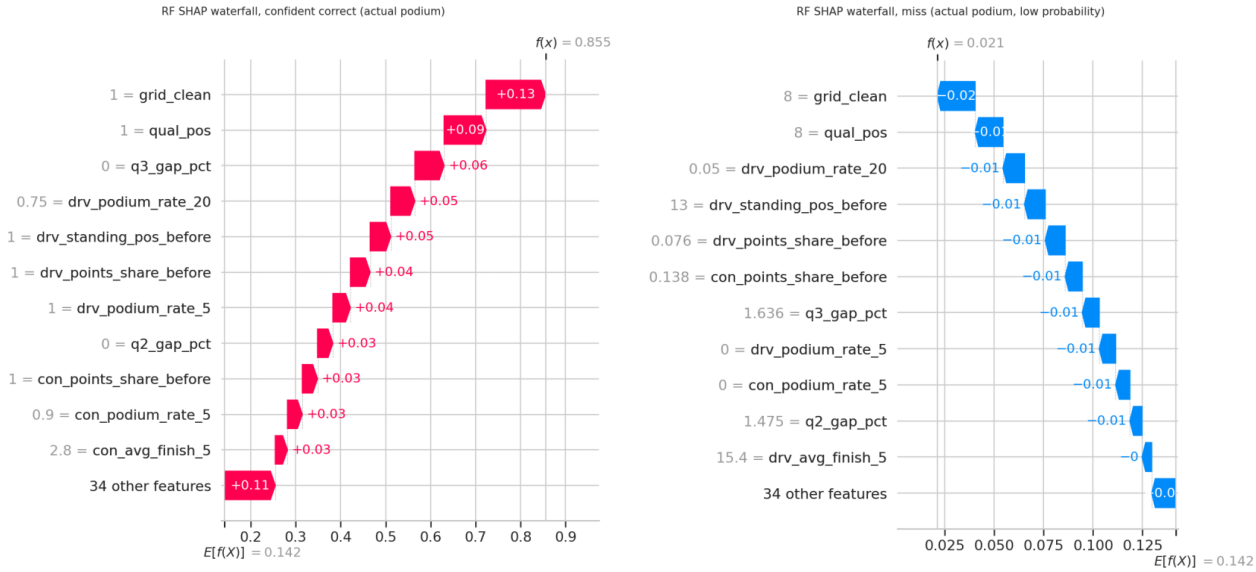


Figure 21. *Random forest SHAP waterfalls. Left: Lewis Hamilton, 2020 Belgian GP (confident and correct, probability 0.855). Right: Esteban Ocon, 2021 Hungarian GP (a podium the model missed, probability 0.021).*

WHAT THE CHART SHOWS

A waterfall starts from the average prediction, E[f(X)] = 0.142, and adds one feature's contribution per bar until it reaches the model output f(x). Red bars push towards a podium and blue bars away from it. For Hamilton the grid slot, qualifying position and Q3 gap raise the probability to 0.855; for Ocon every major feature lowers it, to 0.021.

Hamilton in the 2020 Belgian GP started first with a podium rate of 0.75 over his last 20 races and led the standings: grid (+0.132), qualifying position (+0.093), Q3 gap (+0.069) and driver form (+0.054) push the probability from 0.142 up to 0.855. He won. Ocon in the 2021 Hungarian GP started eighth with weak form (podium rate 0.05) and every large contribution is negative, ending at 0.021. He won after a first-lap pile-up removed several cars ahead of him, an event no pre-race feature can anticipate, so this is a limit of the prediction cutoff and not a fault in the model.

The false alarm is just as informative. Hamilton in the 2020 Italian GP had almost the same pre-race profile as in Belgium and received almost the same probability (0.850), but finished seventh after a stop-and-go penalty for entering a closed pit lane. The inputs were correct; the outcome depended on an event after the cutoff.

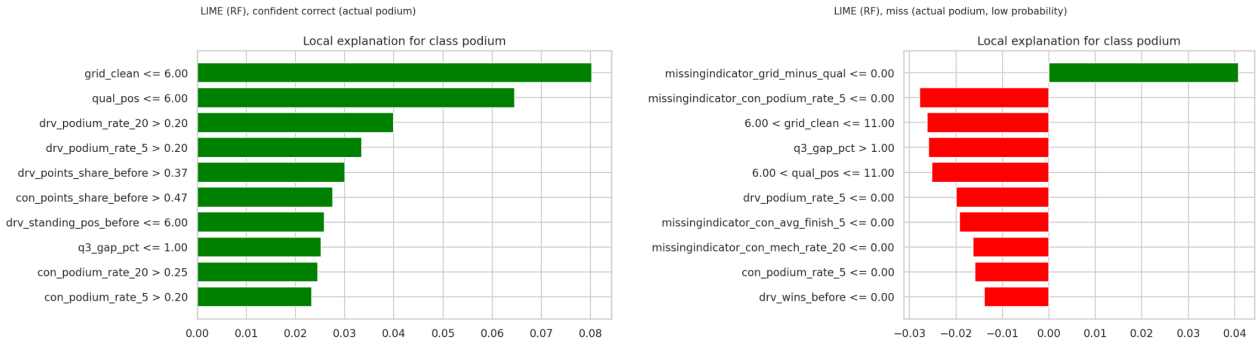


Figure 22. *LIME explanations for the same two cases (forest). Left: Hamilton, 2020 Belgian GP. Right: Ocon, 2021 Hungarian GP.*

WHAT THE CHART SHOWS

Each bar is a condition on one feature (for example grid_clean ≤ 6) with its weight in LIME's local linear model; green bars push towards a podium and red bars away. LIME names the same main drivers as SHAP, but its local model fits loosely: for Hamilton it predicts 0.408 against the forest's 0.855 (R^2 0.66), and for Ocon its fit is poor (R^2 0.10).

LIME broadly agrees with SHAP but not fully: three of the five top features coincide in the confident case and two of five in the miss. Its local surrogate fits only moderately in the confident case (R^2 0.66, local prediction 0.408 against the model's 0.855) and poorly in the miss (R^2 0.10). LIME fits a straight line to random variations of the row, and for an extreme row such as a pole sitter in top form the forest's output lies far above what that line can reach. SHAP is therefore the primary local explanation, since its contributions add exactly to the model output (maximum error 0.0), and LIME is used as a rough check of direction.

8 Inference function

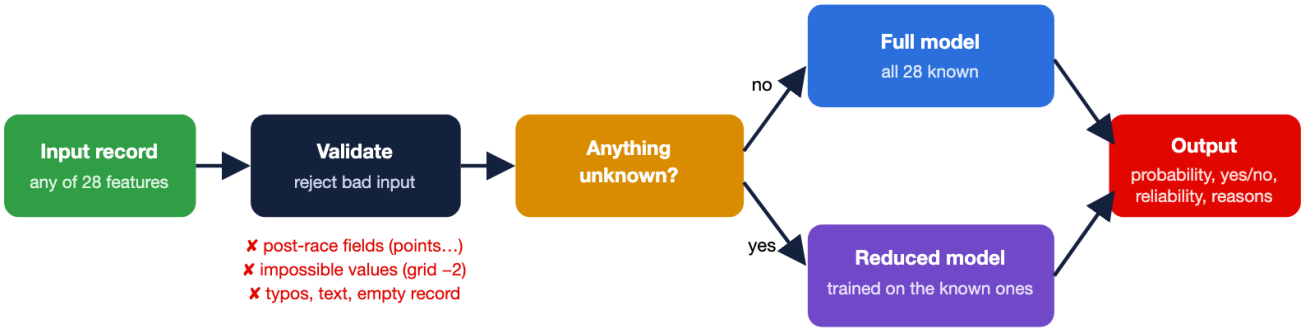


Diagram 7. *How a record flows through PodiumPredictor.*

PodiumPredictor loads the saved preprocessing and models and predicts one record. It validates the input: unknown or misspelled field names, text where a number is expected, impossible values such as a grid slot of -2 or a podium rate of 1.7, inconsistent flags such as reached Q3 without reaching Q2, and an empty record are rejected with a clear message, and post-race fields such as positionOrder or points are refused, which protects the cutoff. A value outside the training range, for example a grid slot of 38 when training saw 1 to 24, is accepted with a warning.

The function separates two kinds of missing input. An unknown input, one the user leaves out or passes as None, is not guessed: the function switches to a reduced model of the same family, trained on the same 2003 to 2019 rows with only the supplied features and the selected hyperparameters, with its own F1 threshold chosen on validation. It reports that model's validation PR-AUC, so the user can see what the missing information costs. A value that is missing as in the data, passed as NaN (for example no Q3 time for a driver knocked out in Q2), goes through the full model exactly as in training. Derived fields (grid_minus_qual, qual_missing) are recomputed. The function returns the probability and decision, which model was used, the largest contributions (exact for the logistic regression and the forest) and a reliability label: low if the grid slot is unknown or missing or more than half the inputs are unknown, medium if a core input (qualifying position or driver form) is unknown or more than 20% are, otherwise high.

Why reduced models. A first version filled unknown inputs with training medians. The models use missing-value flags, and in the training seasons a missing value means something specific (an old qualifying format, a rookie, an unusual entry), so median filling pushed records with unknown inputs into those patterns. For Lando Norris on pole at the 2024 Abu Dhabi GP, the network gave 0.000 when the constructor form was unknown and 0.994 when the driver form was unknown. With only the grid known, the three models gave 0.003 (LR), 0.113 (RF) and 0.000 (NN), against a training podium rate from pole of 80.3%.

Complete record: Norris, Abu Dhabi 2024 (all 28 inputs, started first, finished first). The logistic regression gives 82.2%, the forest 72.0% and the network 96.5%, all above their thresholds, with reliability high; the largest contributions are the grid slot, qualifying position and Q3 gap, as in the global results.

Partially missing records: with 10 inputs unknown (qualifying gaps, driver form, circuit history) the reliability is medium and the reduced models give 71.3% (LR), 71.5% (RF) and 90.3% (NN); their validation PR-AUC is 0.719 to 0.732, against 0.742 to 0.744 for the full models. With only three inputs (grid 5, qualifying position 5, team podium rate 0.4) the reliability is low and the answers are 34.4% (LR) and 23.4% (RF); the forest's prediction rests on qualifying position (+0.072), team form (+0.068) and the grid slot (-0.048). With only the grid slot unknown, the forest gives 68.4%.

Table 10. *Norris at Abu Dhabi 2024: podium probability when one feature group at a time is unknown, first version (median fill) against reduced models.*

Unknown group	LR median fill	LR reduced	RF median fill	RF reduced	NN median fill	NN reduced	Reliability
none (complete)	0.822	0.822	0.720	0.720	0.965	0.965	high
grid	0.504	0.821	0.525	0.691	0.866	0.967	low
qualifying	0.325	0.751	0.437	0.637	0.015	0.962	medium
driver form	0.771	0.774	0.643	0.735	0.994	0.952	medium
constructor form	0.165	0.806	0.646	0.753	0.000	0.961	high
standings	0.525	0.754	0.577	0.706	0.939	0.947	medium
circuit	0.798	0.808	0.715	0.730	0.967	0.953	high
era	0.822	0.827	0.720	0.710	0.965	0.951	high
everything except grid	0.003	0.676	0.113	0.803	0.000	0.941	low

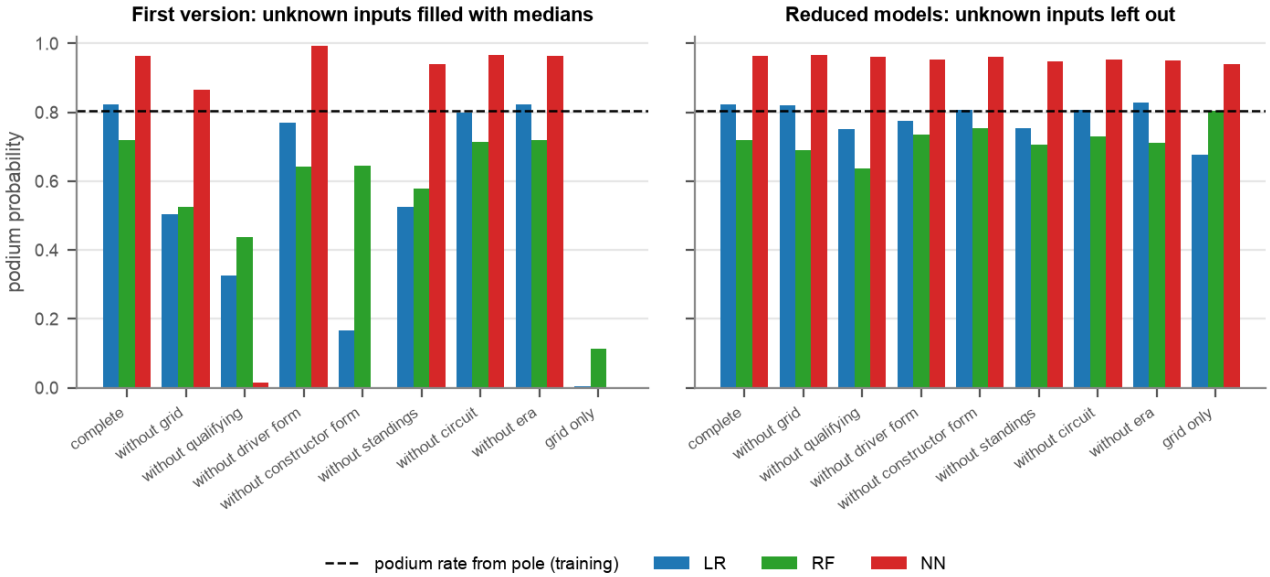


Figure 23. *Norris at Abu Dhabi 2024: predicted podium probability as one feature group at a time is unknown. Left: the first version, which filled unknown inputs with training medians. Right: reduced models trained on the supplied features. The dashed line is the 80.3% training podium rate from pole.*

WHAT THE CHART SHOWS

Each group of bars is one scenario for Norris on pole at Abu Dhabi 2024, with one feature group unknown; the dashed line is the 80.3% training podium rate from pole. With median filling (left), predictions swing from 0.000 to 0.994 depending on what is missing. With reduced models (right), they stay close to the reference line: the forest follows it most closely (0.803 with only the grid known), while the network is consistently over-confident.

With reduced models the predictions stay in a narrow band: 0.68 to 0.83 for the logistic regression, 0.64 to 0.80 for the forest and 0.94 to 0.97 for the network, against 0.003 to 0.822, 0.113 to 0.720 and 0.000 to 0.994 with median filling. With only the grid known, the reduced forest gives 0.803, matching the 80.3% training podium rate from pole. The reduced logistic regression gives 0.676, because a single linear term in the grid slot cannot capture how steep the advantage of pole is, and the network gives 0.941, consistent with its class-weighted, uncalibrated probabilities. Reduced forests consider every column at each split when fewer than ten columns remain: an earlier version that used the square root of the column count left about half of its trees unable to split and gave the pole sitter 0.49. The notebook counts trees that never split as a check, and the count is 0 of 300 in every scenario. Reduced models cost accuracy in proportion to what is missing: the forest's validation PR-AUC falls from 0.744 with all inputs to between 0.719 and 0.746 with one feature group unknown, and to 0.649 with only the grid.

The function also reproduces the stored predictions of Section 5 from disk (maximum difference 2e-08), and reduced models built from the saved files give identical results, so it can be used independently of the notebook session.

9 Known limitations

- **Qualifying coverage drift.** Qualifying data are absent before 1994 and partial until 2002 (Figure 3). Models train from 2003 only, so the models cannot learn from earlier seasons, and the history serves only the form features, which use race results alone.
- **Era sensitivity.** The value of a grid slot changed with regulations (Figure 8), the test seasons follow new rules, and the era flags cannot be evaluated, because they are constant on the validation and test seasons and the 2022 rules have no flag of their own, so adaptation to future rule changes is not guaranteed. Section 3.1 shows that circuit rankings also shift between eras.
- **Small validation set.** Validation holds 39 races and 775 rows; the test set has 68 races. Confidence intervals are wide, the three models are statistically tied, each selected attempt is tied with most alternatives in its family (Section 5.2), and thresholds tuned on validation transfer imperfectly (test F1 below validation F1).
- **A strong baseline.** A one-line rule reaches test PR-AUC 0.647, only 0.05 to 0.07 below the learned models, whose intervals are about 0.1 wide. The models add information mostly through qualifying gaps and driver form, and an improvement beyond about 0.72 would need information we exclude by design, such as weather, tyre strategy and race-day events.
- **Collinearity and ablation limits.** Grid and qualifying position (0.97) and driver and team standings (above 0.9) overlap, so importance and ablation shares are partly arbitrary. The circuit group hurt the logistic regression and was kept in the delivered models.
- **Team identity drift.** Lineage keys exist only from 1999, so before 1999 a constructor rename resets that team's form history, and drivers who change teams mid-season carry only their personal form.
- **Manual mappings.** Team lineage keys (from 1999), the nationality-to-country map for the home analysis, the status grouping for mechanical retirements and the status treatment of 'Retired' are judgement calls that were reviewed but are not unique. Counting the generic Retired status (with Fuel leak) as mechanical leaves the Q3 rankings unchanged (Section 3.3).
- **Cutoff and rule R2.** Rule R2 removes 47 rows from 2003 on in which a driver qualified but did not start, which uses information from just after the prediction cutoff. All 47 rows are non-podiums and the target is undefined for a car that never races, but a deployed model would still have to score such drivers.
- **Model behaviour.** The neural network's probabilities are uncalibrated (class weighting) and its SHAP values are approximate (150 rows). LIME fits poorly for extreme rows (R^2 0.10 in one case). Reduced models for unknown inputs keep predictions stable but cost up to about 0.10 validation PR-AUC when most inputs are unknown, and callers must pass None for unknown values and NaN only for values missing as in the data. Drivers and teams new to the data have thin history.
- **No causal claim.** Explanations describe the models. The home-race and circuit findings are also associations in observational data.

Decision	Basis	Evidence	Status
Train from 2003, history only for form features	Hypothesis and trial	Coverage by season (Figure 3); Experiment C shows no gain from earlier training years	Supported
Chronological split at 2019 and 2021	Reasoned	A random split would let later seasons inform earlier ones and put drivers of one race on both sides; validation is the smallest period with enough podiums to tune thresholds; the test seasons are the unseen 2022 rules (Section 2.3)	Supported
R1 drop Indianapolis 500 races	Hypothesis	Brief and guide describe the distortion; the effect on the home-race result was not measured	Partly
R2 drop non-starters	Hypothesis	grid = 0 breakdown and the cleaning-effect figure (Figure 2); 47 rows from 2003 on use information from just after the cutoff, documented in Table 1 and Section 9	Supported
R3 keep non-zero grid, lowest grid, lowest resultId for shared drives	Reasoned	61 pairs, 17 where a dropped row was a podium; alternative rules not compared. All pairs are before 1970, outside the training window, so the models are unaffected	Partly
R4 pit-lane flag and last slot plus	Hypothesis	grid = 0 by decade; 0 podiums in 73 modern pit-lane starts	Supported
R5 cap qualifying laps at 1.5 times the race median	Hypothesis	Quantiles of lap against race median: one lap affected	Supported
Feature groups: grid, qualifying, form, standings	Hypothesis and trial	Single-feature screen, Figure 9, ablation (Section 6)	Supported
Keep features with no single-feature signal	Hypothesis	Table 3a: no pit-lane starter on the podium; penalised drivers outperform their grid slot (7.3% against 2.8%); standings are empty at season openers	Supported
shift(1) for form, as-of standings	Trial	Future-invariance test and Experiment E (leaky variants reach 0.97 to 0.999)	Supported
Rolling windows of 5 and 20 races, at least 3 past races	Trial	Experiment F: windows of 3 and 10 and of 10 and 40 races change validation ROC-AUC by -0.0007 and -0.0003, within noise	Supported
Points as share of leader, not raw points	Trial	Experiment F: raw points +0.0007, within noise; the share is kept for comparability across the 2010 scale change	Supported
Qualifying gap in percent, not raw seconds	Trial	Experiment F: raw seconds -0.0002, within noise; the percent gap is kept for comparability across tracks	Supported
Era flags	Hypothesis	Figure 8 and Table 3a: pole podium rate 79.0%, 77.7% and 83.5% across the three training eras; the flags are constant on validation and test, so their contribution cannot be measured there	Partly
Lineage key for team form	Trial	Experiment D2: team form on the raw constructorId lowers validation ROC-AUC by 0.0005 (CI -0.0009 to -0.0000); Experiment D: identity columns add nothing	Supported
Circuit group	Hypothesis, then trial	Weak screen (AUC 0.704); removal improves the logistic regression by +0.011 but not the forest or the network; models were fixed before the ablation	Contradicted for LR, kept
Reject home flag and circuit conversion rate	Hypothesis	Questions 1 and 2: no detectable effect	Supported
Reject sprint results	Reasoned	Available only from 2021, so empty for nearly all training rows	Partly
Keep correlated features	Reasoned and trial	Regularised models; ablation tests each group	Partly
Fit preprocessing on training rows only	Trial	Experiment A: negligible difference, kept as the correct protocol	Supported
Median imputation	Trial	Experiment B against zero fill and mean fill	Supported
Missing-value flags	Hypothesis and trial	Missingness is structured by season and format (Figure 3); Experiment F: flags on +0.0026 (CI -0.0001 to	Supported
Standard scaling	Trial	Experiment F: no scaling -0.0003 for the logistic regression, within noise; kept because the network needs scaled inputs	Supported

Decision	Basis	Evidence	Status
Model families and baseline	Trial	Table 7: nine attempts with before and after	Supported
Hyperparameters	Trial	C sweep and forest grid (Section 5.2)	Supported
Selection among tied attempts	Trial and reasoned	Bootstrap tie check (Section 5.2): only A4 against A3 is reliable; the other selections rest on design	Partly
Class weights	Trial	A2, A5 and A8 against their parents	Supported
Network size and regularisation	Trial	A6 against A7 and A8	Supported
Threshold: F1 maximum on validation	Reasoned	F1 is the brief's threshold metric; other criteria not compared	Partly
PR-AUC as primary metric, race-level bootstrap	Reasoned	About 15% positives; drivers within a race are dependent	Supported
Reduced models for unknown inputs	Trial	Section 8 and Table 10: for the same pole sitter, median filling gives 0.000 to 0.994 across nine scenarios; reduced models stay between 0.64 and 0.97	Supported
Every column per split in reduced forests with fewer than 10 columns	Trial	With the square root of the column count, about half the trees never split and the pole sitter gets 0.49; with every column, 0.803 and no unsplit trees	Supported
Reliability cut-offs of 50% and 20% unknown	Reasoned	Not tuned; each reduced model's validation PR-AUC is reported as the direct measure of what is lost	Not tested
Random forest for partial records	Trial	Grid only: reduced forest 0.803 against the 80.3% pole rate; reduced LR 0.676 (underestimates), reduced NN 0.941 (overstates)	Supported

Experiments D2 and F turned five reasoned choices into tested ones: the rolling windows, points share, percent gaps, scaling and the missing-value flags. None of the alternatives beat the chosen setting beyond noise, and the lineage key showed a small, consistent gain. Only the reliability cut-offs of the inference function are still not compared with an alternative. Several decisions remain supported only indirectly: R1, R3, the era flags (which cannot be evaluated on the 2014 onward seasons), the sprint exclusion, the correlated features and the F1 threshold rule.